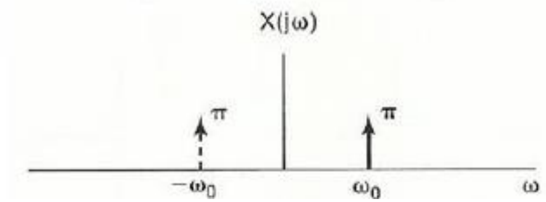
- Homework: 8.4, 8.25, 8.29
- Tutorial: 8.34, 8.35, 8.38

Aliasing: Example

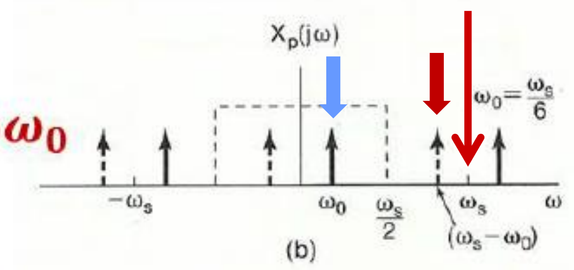
Signal before sampling: $\cos \omega_0 t$

Sampling rate: ω_s

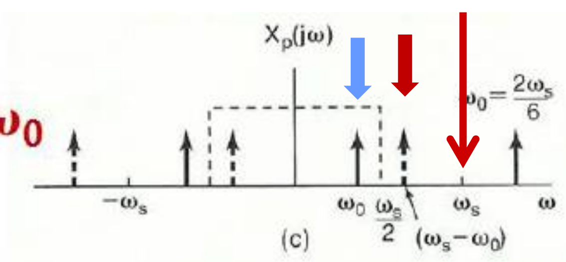


Lowpass Filter: $\dfrac{-\omega_s}{2} \sim \dfrac{\omega_s}{2}$

$\omega_s = 6\omega_0$

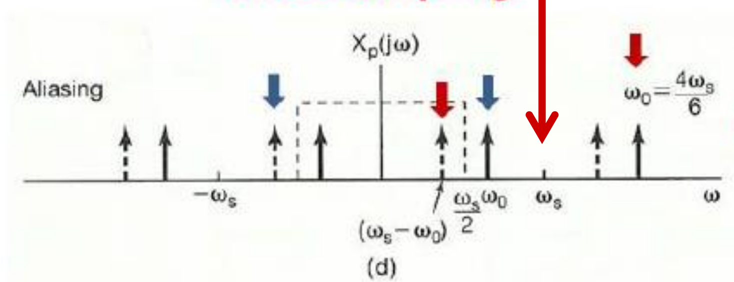


$\omega_s = 3\omega_0$

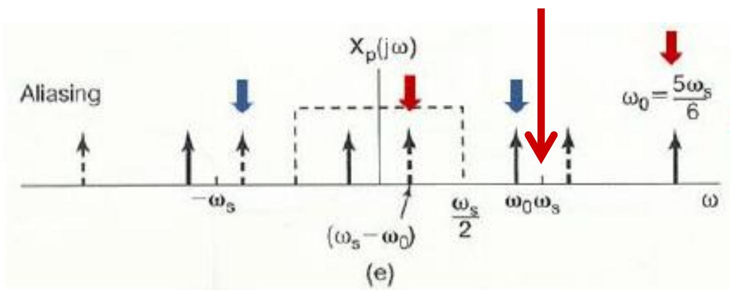


$\cos \omega_0 t$

Undersampling



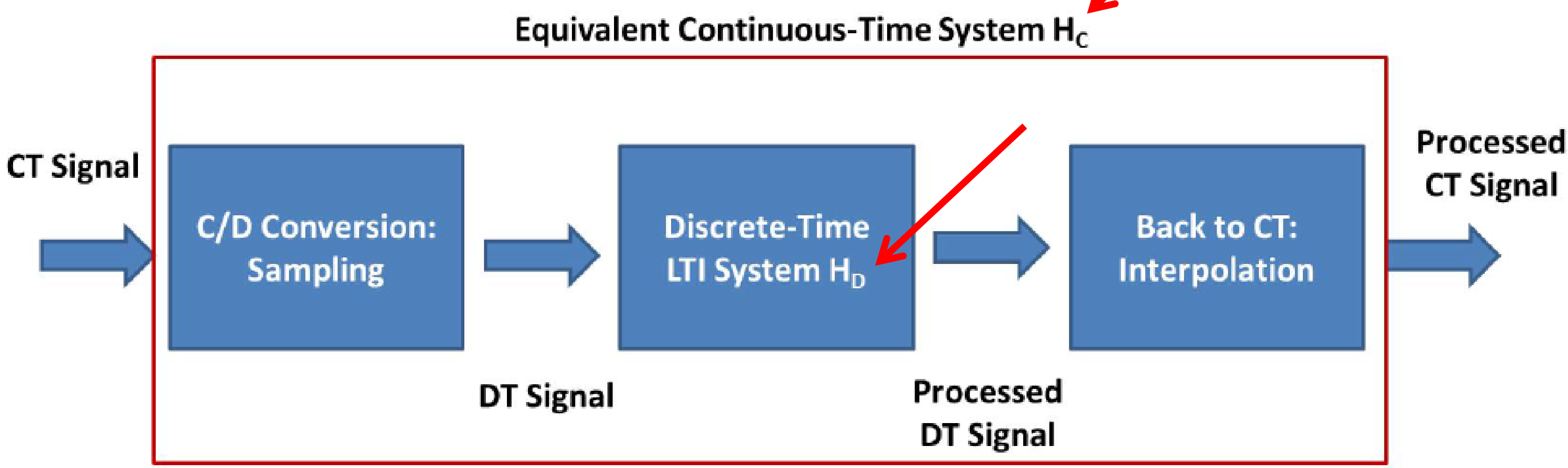
$\omega_s = \frac{3}{2}\omega_0$



$\omega_s = \frac{6}{5}\omega_0$

Aliasing: $\cos(\omega_s - \omega_0)t$

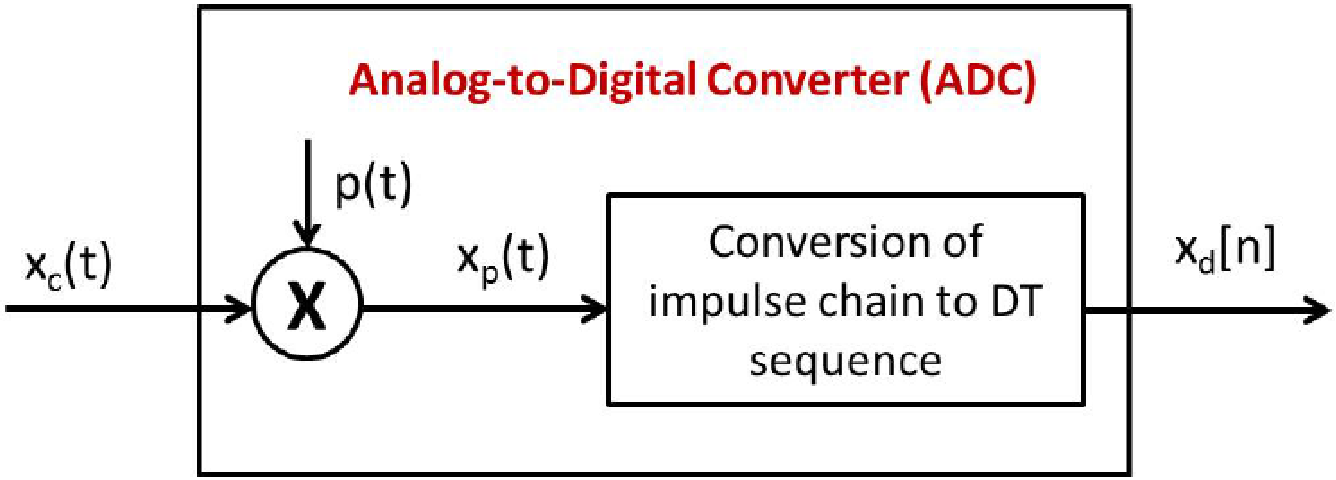
Block Diagram



- It is much easier to design DT system.
- What's the relation between H_C and H_D?

$$H_C = \tilde{H}_D$$

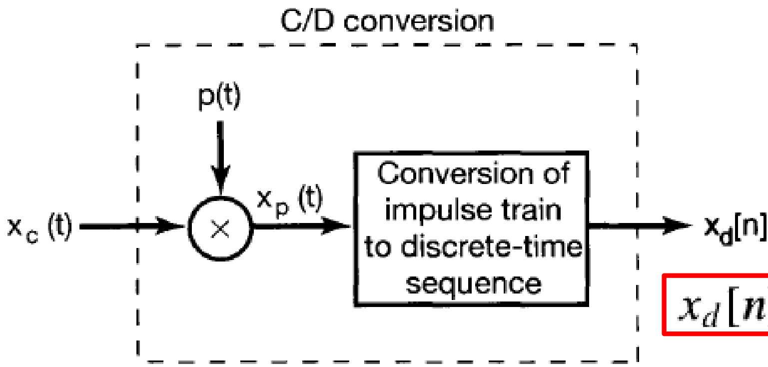
Discretization: C/D Conversion



- Mathematical Interpretation (Fourier Transform)

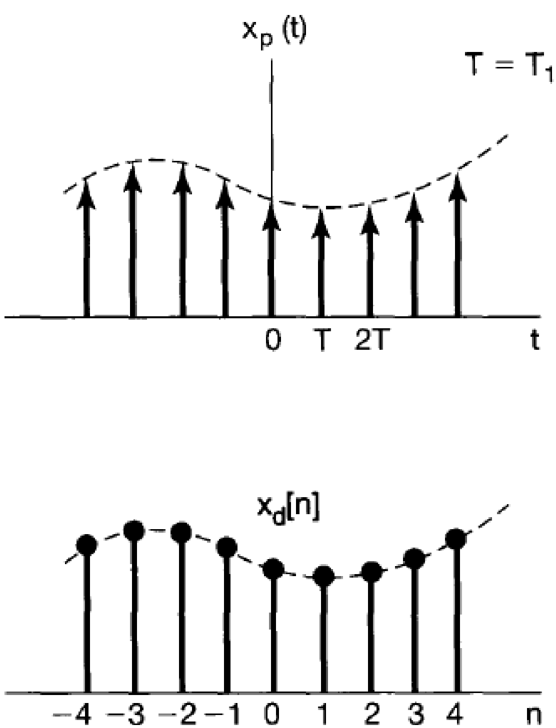
$$x_c(t) \quad \longleftrightarrow \quad X_c(j\omega)$$

$$x_p(t) = \sum_{n=-\infty}^{+\infty} x_c(nT)\delta(t - nT) \quad \longleftrightarrow \quad X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s))$$



$$\boxed{x_d[n] = x_c(nT)}$$

(a)

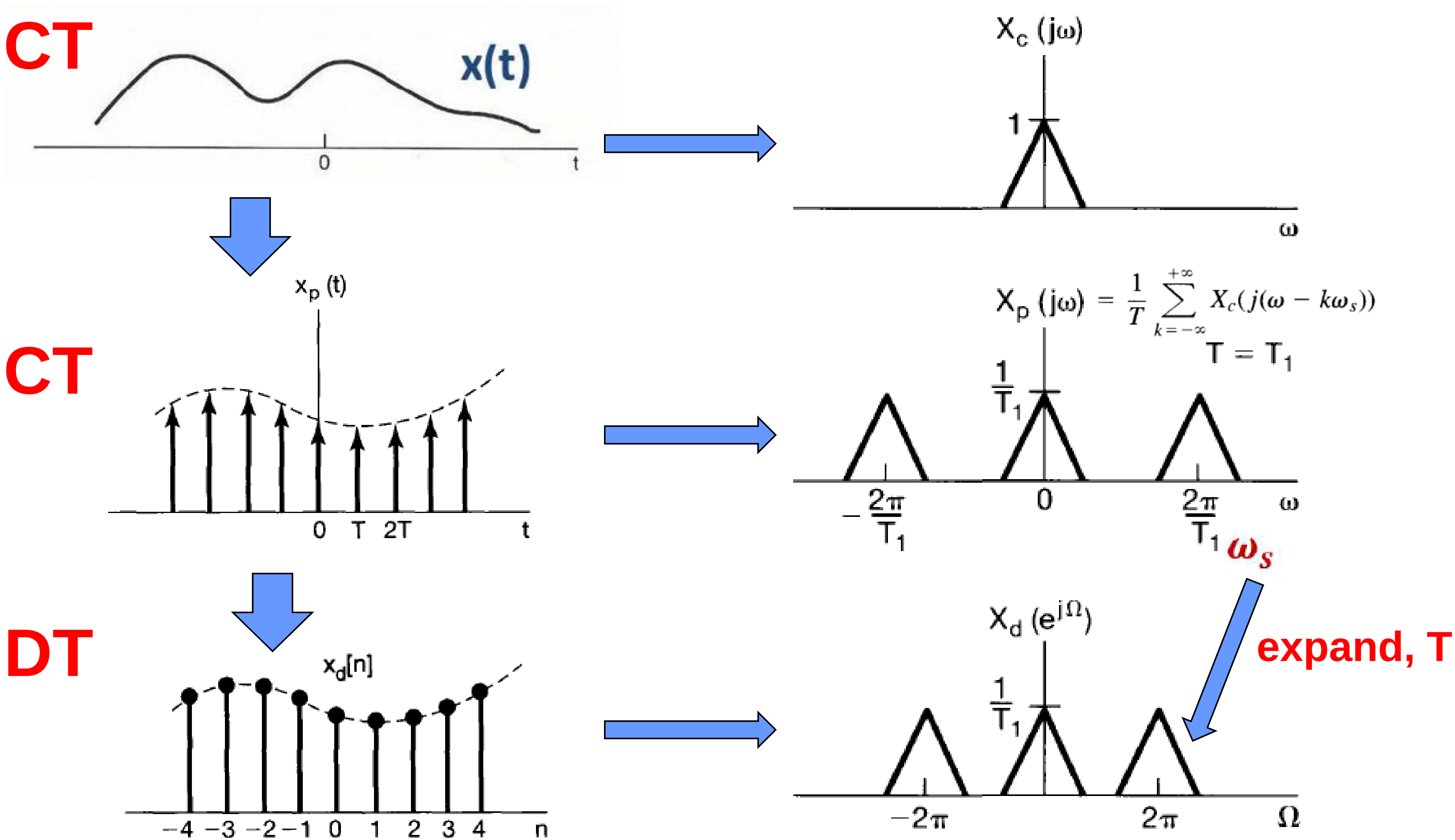


$$X_d(e^{j\Omega}) = \sum_{n=-\infty}^{+\infty} x_d[n]e^{-j\Omega n}$$

$$= \sum_{n=-\infty}^{+\infty} x_c(nT)e^{-j\Omega n}$$

$$X_p(j\omega) = \sum_{n=-\infty}^{+\infty} x_c(nT)e^{-j\omega nT}$$

$$X_d(e^{j\Omega}) = X_p(j\Omega/T)$$



$$X_d(e^{j\Omega}) = X_p(j\Omega/T)$$

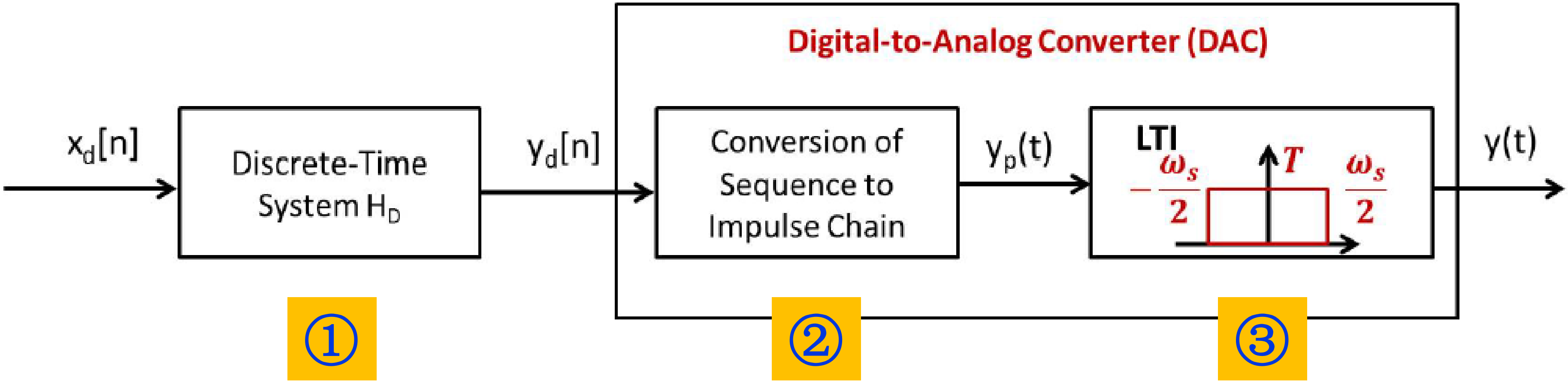
$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s))$$

$$X_d(e^{j\omega}) = X_p(j\omega / T)$$

$$X_d(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega}{T} - k\omega_s))$$

$$= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega - 2\pi k}{T}))$$

DT Processing and Conversion



- Mathematical Interpretation (Fourier Transform)

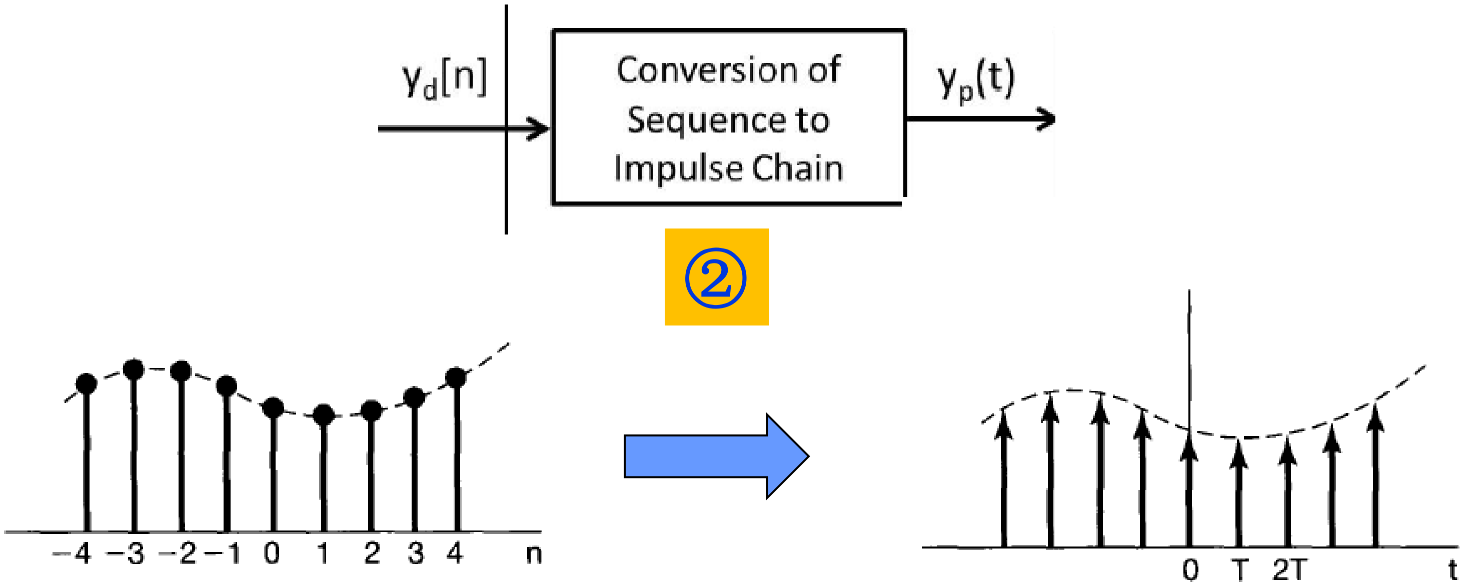
① $$y_d[n] = x_d[n] * h_D[n] \quad \longleftrightarrow \quad Y_d(e^{j\omega}) = X_d(e^{j\omega})H_D(e^{j\omega})$$

② $$y_p(t) = \sum_{n=-\infty}^{\infty} y_d[n]\delta(t - nT) \quad \longleftrightarrow \quad Y_p(j\omega) = Y_d(e^{j\omega T})$$

③ $$y(t) = y_p(t) * h_{LP}(t) \quad \longleftrightarrow \quad Y(j\omega) = Y_p(j\omega)H_{LP}(j\omega)$$

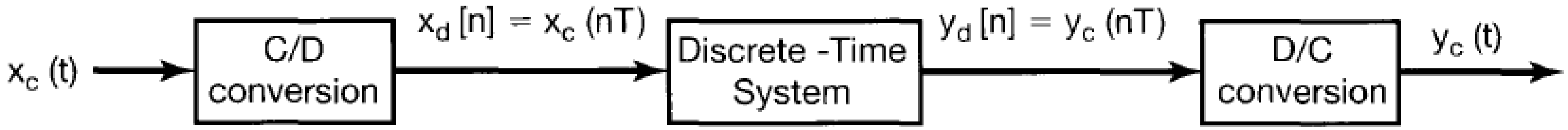
$$Y(j\omega) = X_d(e^{j\omega T})H_D(e^{j\omega T})H_{LP}(j\omega)$$

$y_d[n] = y_c(nT)$



1. **n → t**
2. **time-scaling**

$$y_p(t) = \sum_{n=-\infty}^{\infty} y_d[n]\delta(t - nT) \quad \longleftrightarrow \quad Y_p(j\omega) = Y_d(e^{j\omega T})$$



$$Y(j\omega) = X_d(e^{j\omega T}) H_D(e^{j\omega T}) H_{LP}(j\omega)$$

$$X_d(e^{j\omega}) = \frac{1}{T}\sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega}{T} - k\omega_s\right)\right)$$

幅值为T

$-\frac{\omega_s}{2} \sim \frac{\omega_s}{2}$

$$\begin{aligned} Y(j\omega) &= \left[\frac{1}{T}\sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s))\right] H_D(e^{j\omega T}) H_{LP}(j\omega) \\ &= X_c(j\omega) H_D(e^{j\omega T}) \\ &= X_c(j\omega)\widetilde{H}_D(e^{j\omega T}) \end{aligned} \quad (1)$$

$\rightarrow$ 周期

$H_C = \frac{Y}{X_C} = \widetilde{H}_D$

where

$$\widetilde{H}_D(e^{j\omega T}) = \begin{cases} H_D(e^{j\omega T}) & |\omega| < \omega_s/2 \\ 0 & otherwise \end{cases} \quad (2)$$

已知H_D求H_C，先压缩再截取

- It is equivalent to a continuous-time LTI system $H_C(j\omega) = \widetilde{H}_D(e^{j\omega T})$
- $H_D(e^{j\omega T})$ is a periodic extension of $\widetilde{H}_D(e^{j\omega T})$ with period $\omega_s = 2\pi/T$

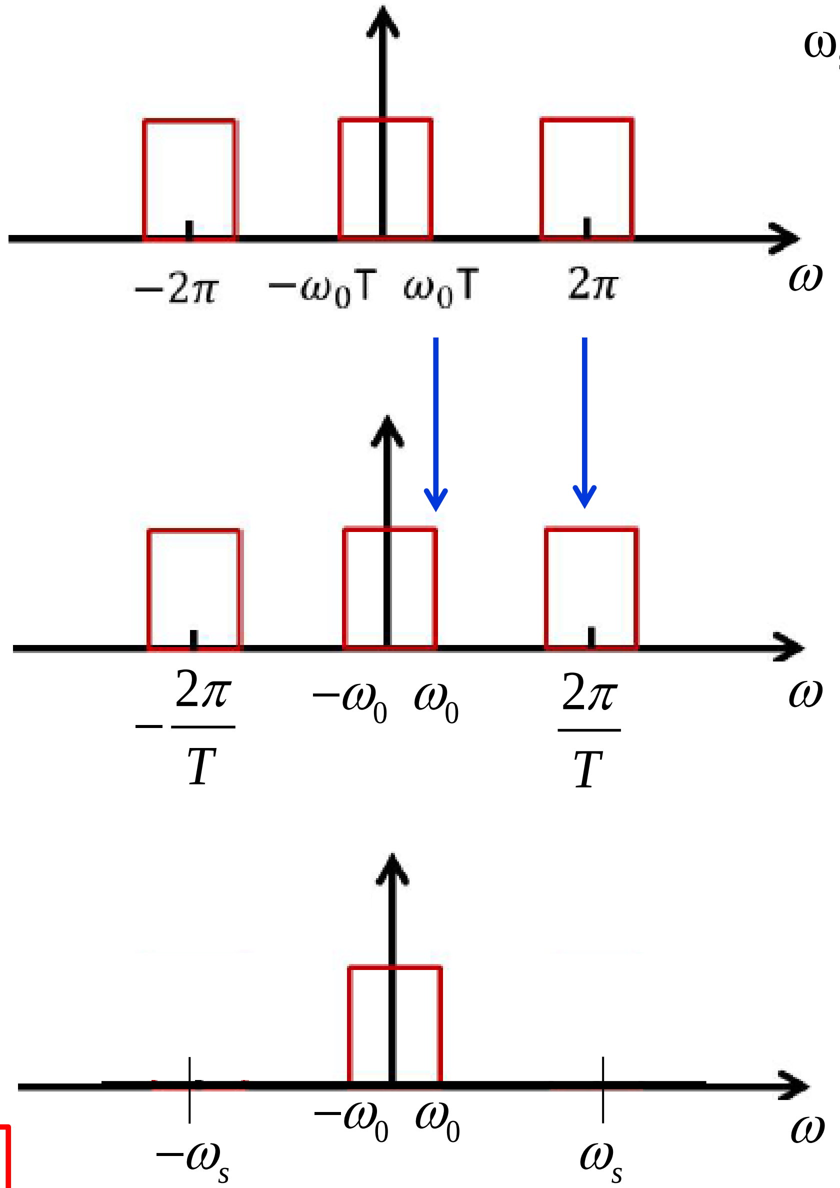
T: sampling period

$\omega_s=2\pi/T$: sampling rate

$H_D(e^{j\omega})$

$H_D(e^{j\omega T})$

$\widetilde{H}_D(e^{j\omega T})$



$$\boxed{H_C(j\omega) = \widetilde{H}_D(e^{j\omega T})}$$

System Design

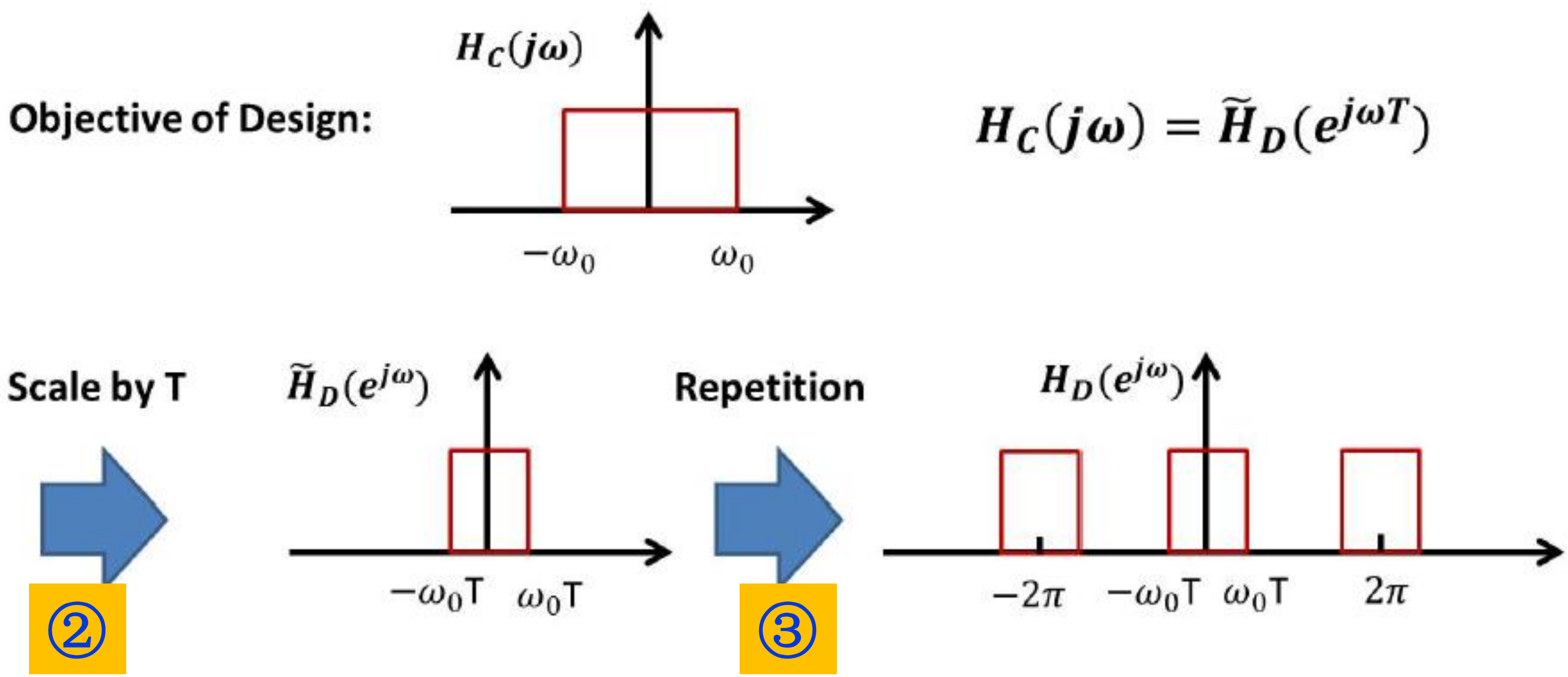
$$H_C(j\omega) = \widetilde{H}_D(e^{j\omega T})$$

- How can we design a CT LTI system with frequency response H_C via DT LTI system?
- Step 1: Sampling frequency ω_s or $2\pi/T$ should be larger than Nyquist rate

  ω_m
- Step 2: $\widetilde{H}_D(e^{j\omega T}) = H_C(j\omega)$
- Step 3: Frequency response of DT LTI system
  $H_D(e^{j\omega T}) = \sum_{k=-\infty}^{\infty} \widetilde{H}_D(e^{j(\omega - k\omega_s)T})$ or
  $H_D(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \widetilde{H}_D(e^{j(\omega - k\omega_s T)}) = \sum_{k=-\infty}^{\infty} H_C(j\frac{\omega - 2k\pi}{T})$

System Design Example

- How to implement an ideal CT lowpass filter?

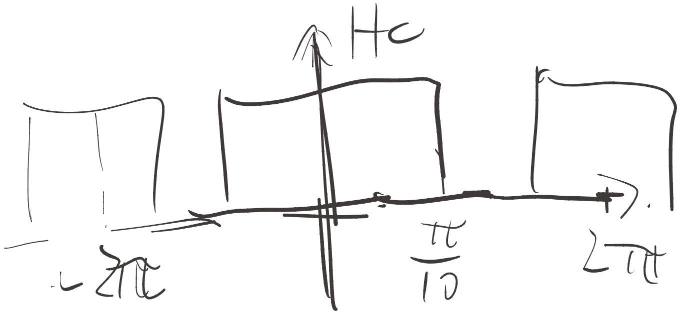


Practice

- Design a DT system to process a CT signal with ideal low-pass filtering
  - cutoff frequency: fcut=50 Hz $\omega_c = 100\pi$
  - Sampling frequency: fs=1000 Hz $\omega_s = 2000\pi$

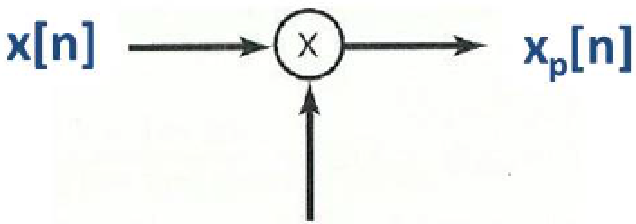
$T = \frac{1}{1000}$ s

$\omega_c \cdot T = \frac{\pi}{10}$



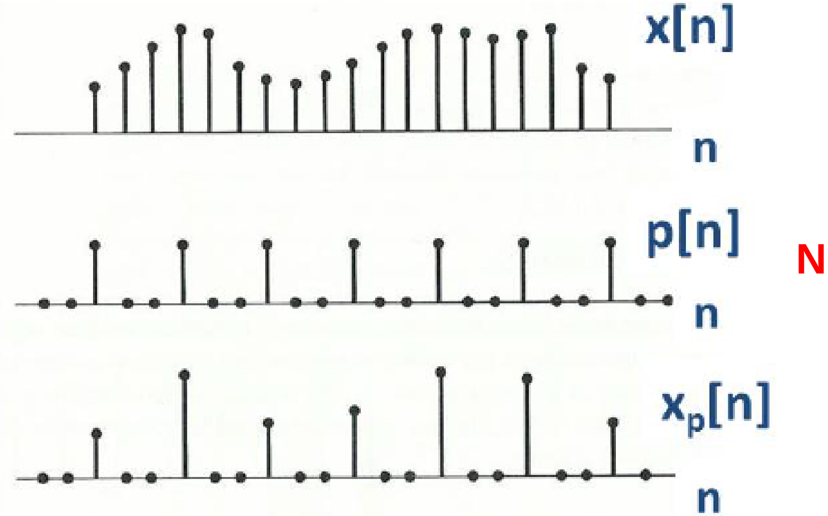
Sampling on Discrete-Time Signals

- In digital signal processing, we may sample the discrete-time signals
- Example: image, audio, video compression
- Its mathematical model is give below:



$$\mathbf{p[n]} = \sum_{\mathbf{k}=-\infty}^{+\infty} \boldsymbol{\delta}[\mathbf{n}-\mathbf{kN}]$$

$$\mathbf{x_p[n]} = \sum_{\mathbf{k}=-\infty}^{+\infty} x[kN]\boldsymbol{\delta}[\mathbf{n}-\mathbf{kN}]$$



Frequency Analysis

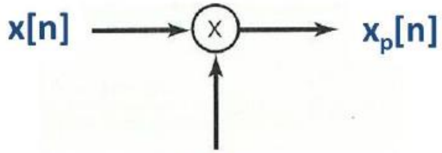
$$P(e^{j\omega}) = \frac{2\pi}{N}\sum_{k=-\infty}^{\infty}\delta(\omega - k\omega_s) \quad where \boxed{\omega_s = 2\pi/N}$$

$$X_p(e^{j\omega}) = \frac{1}{2\pi}\int_{2\pi} P(e^{j\theta})X(e^{j(\omega-\theta)})d\theta = \boxed{\frac{1}{N}}\sum_{k=0}^{N-1} X(e^{j(\omega-k\omega_s)})$$

periodic convolution

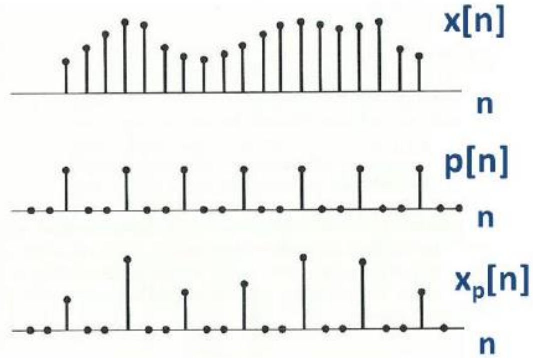
$$= \frac{1}{N}[X(e^{j\omega}) + X(e^{j(\omega-\omega_s)}) + \ldots + X(e^{j(\omega-(N-1)\omega_s)})]$$

What is the relation between $X(e^{j\omega})$ and $X_p(e^{j\omega})$?



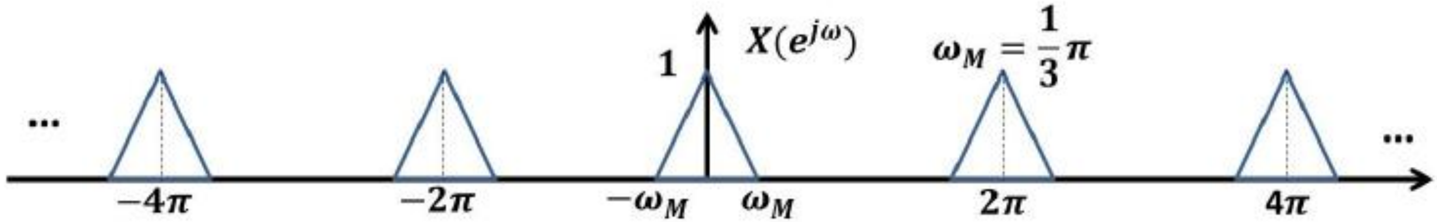
$$p[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$$

$$x_p[n] = \sum_{k=-\infty}^{+\infty} x[kN]\delta[n - kN]$$



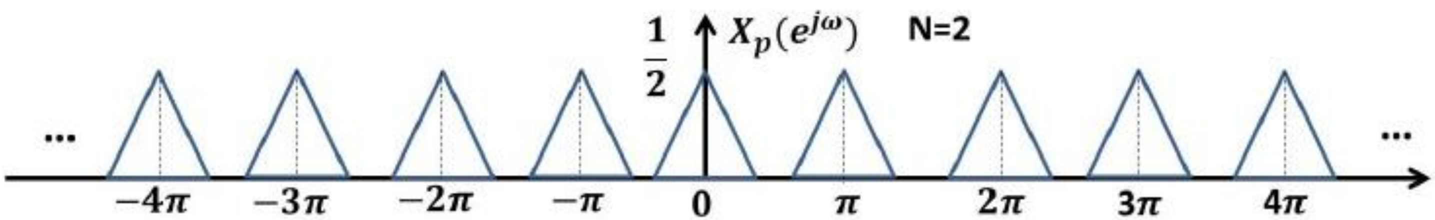
N

$\boxed{\omega_s = 2\pi}$



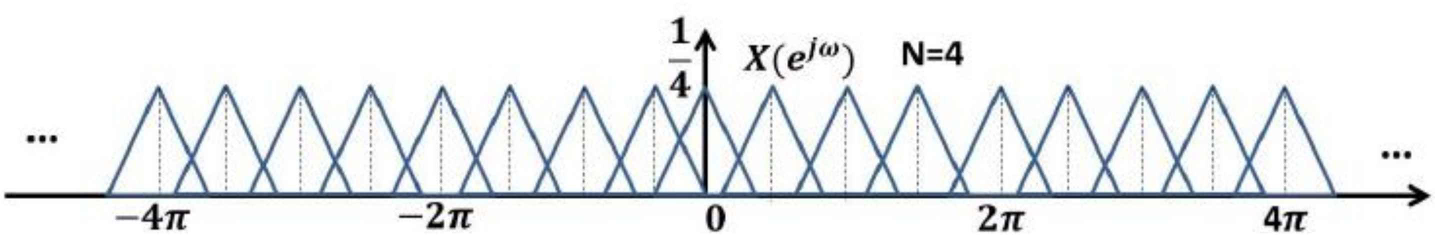
N=2

$\boxed{\omega_s = \pi}$



N=4

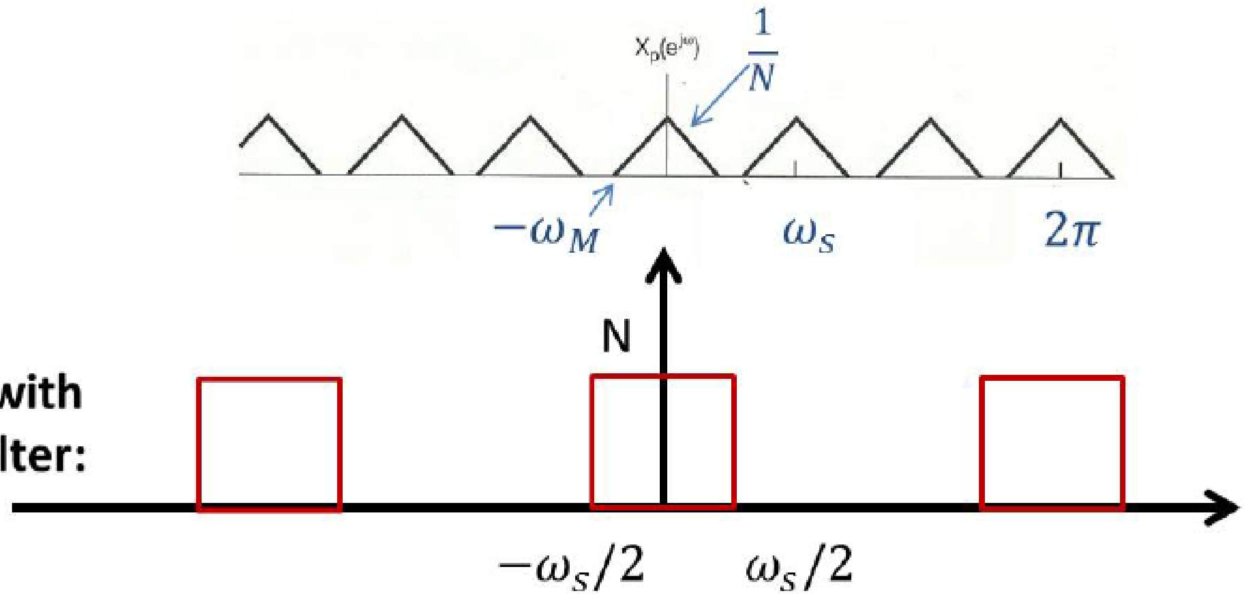
$\boxed{\omega_s = \frac{\pi}{2}}$



Reconstruction

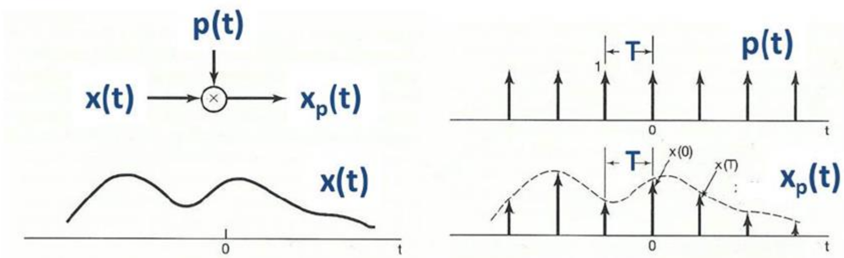
- Perfect reconstruction is applicable when $\omega_s > 2\omega_M \leftrightarrow N < \frac{\pi}{\omega_M}$

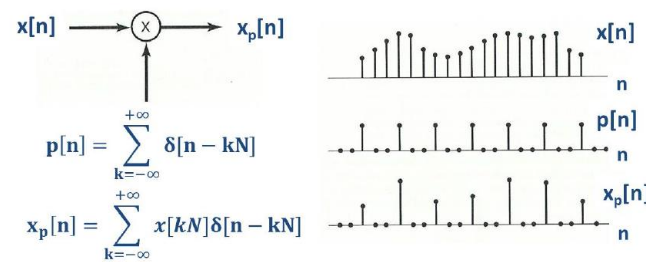
Reconstruction with ideal low-pass filter:



- Aliasing occurs when $\omega_s < 2\omega_M$

Comparison btw CT sampling and DT sampling





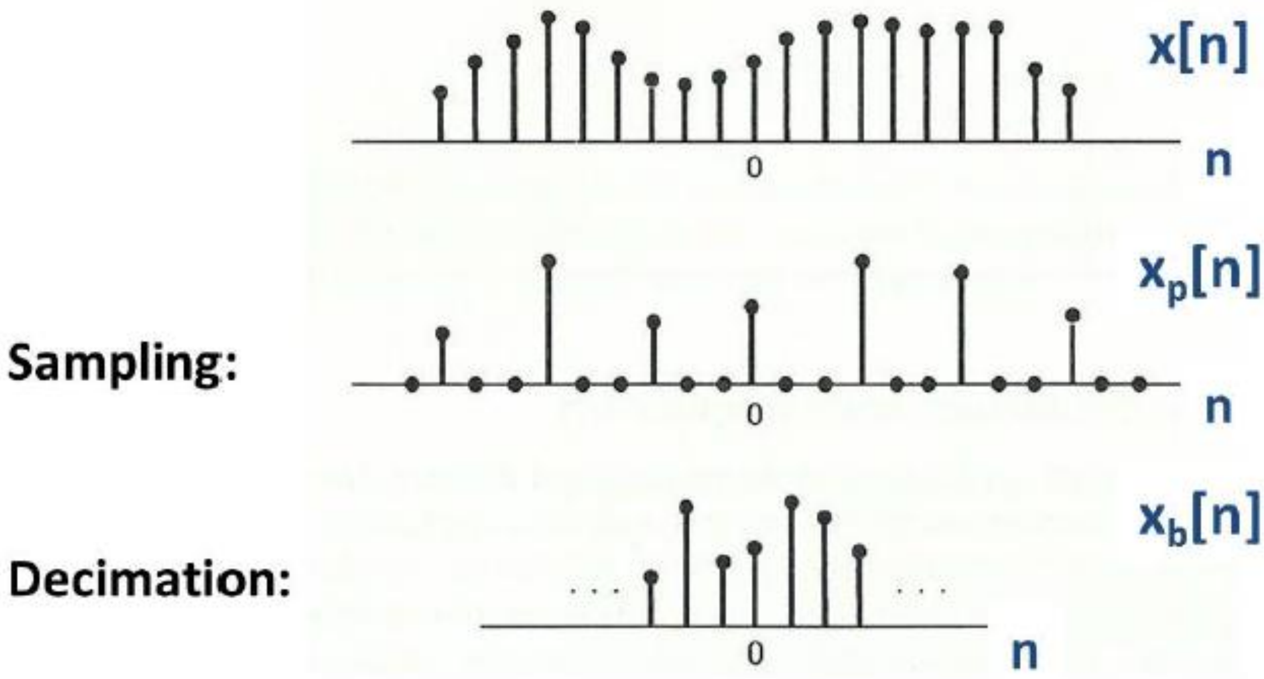
N

Both have sampling theory

	CT	DT
Sampling period/frequency	T, $\omega_s=2\pi/T$	N, $\omega_s=2\pi/N$
Sampling theory	$\omega_s>2\omega_M$	$\omega_s>2\omega_M$
Reconstruction with ideal LPF	$[-\omega_s/2 \quad \omega_s/2]$, gain T	$[-\omega_s/2 \quad \omega_s/2]$, gain N

Decimation

- After sampling, there will be a great amount of redundancy
- Decimation: discrete-time sampling + remove zeros



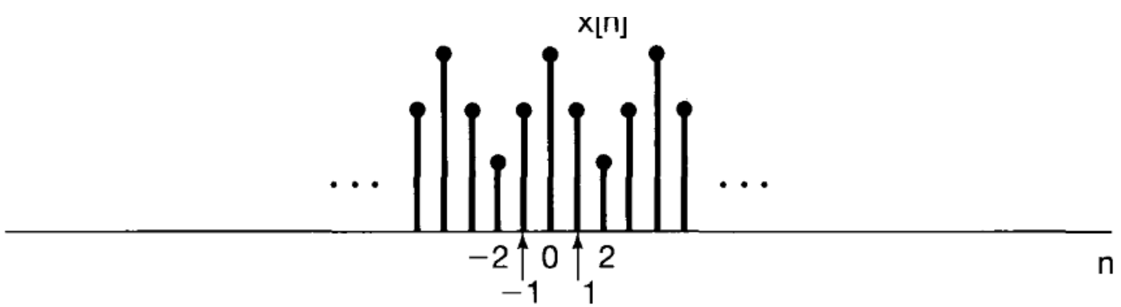
$$x_b[n] = x_p[nN] = x[nN]$$

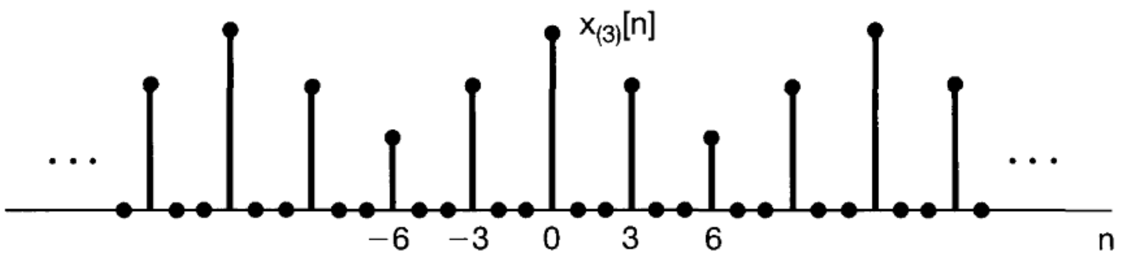
What is the relation between $X(e^{j\omega})$ and $X_b(e^{j\omega})$?

Review

- Time Expansion
  - Define $x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of k} \\ 0, & \text{Otherwise} \end{cases}$, then

$$x_{(k)}[n] \longleftrightarrow X(e^{jk\omega})$$





Review

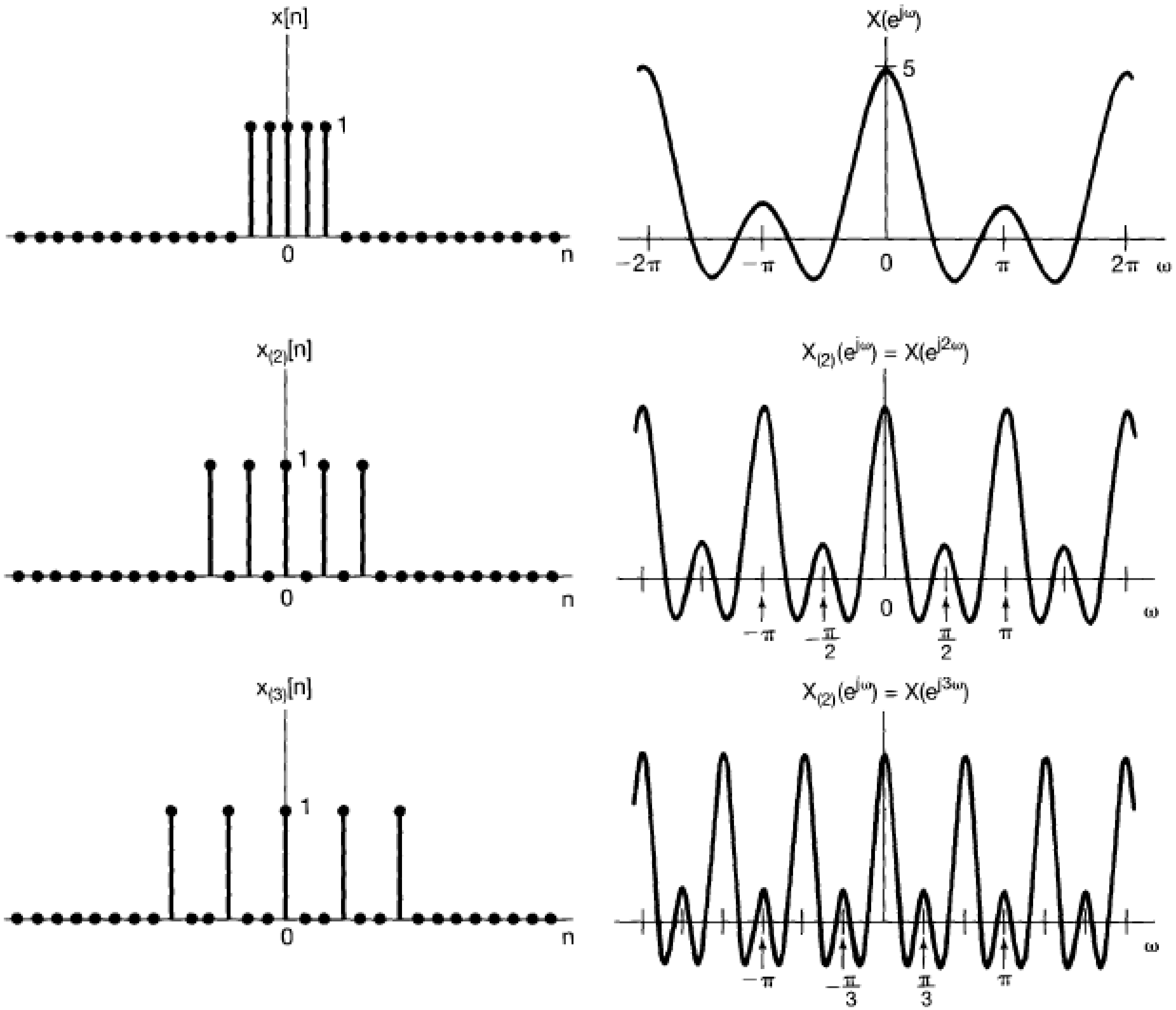


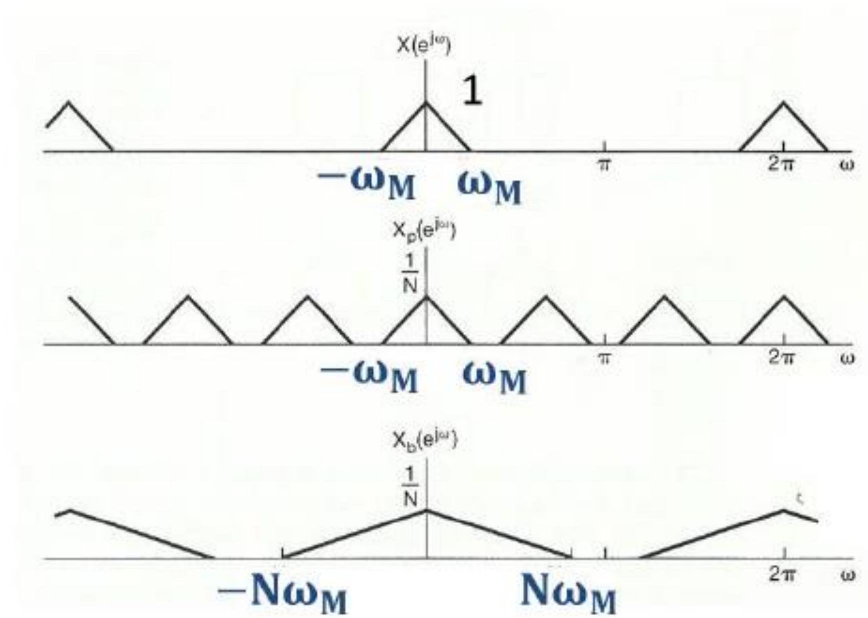
Figure 5.14 Inverse relationship between the time and frequency domains: As k increases, $x_{(k)}[n]$ spreads out while its transform is compressed.

Frequency Analysis

$$
\begin{aligned}
X_b(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x_b[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x_p[n] e^{-j\omega n/N} = X_p(e^{j\omega/N}) \\
&= \frac{1}{N}\sum_{k=0}^{N-1} X(e^{j\omega/N - k\omega_s})
\end{aligned}
$$

$$\boxed{X_p(e^{j\omega}) = \frac{1}{N}\sum_{k=0}^{N-1} X(e^{j(\omega-k\omega_s)})}$$

$$\boxed{\omega_s = 2\pi/N}$$



$$\boxed{\omega_s = 2\pi}$$

Sampling:

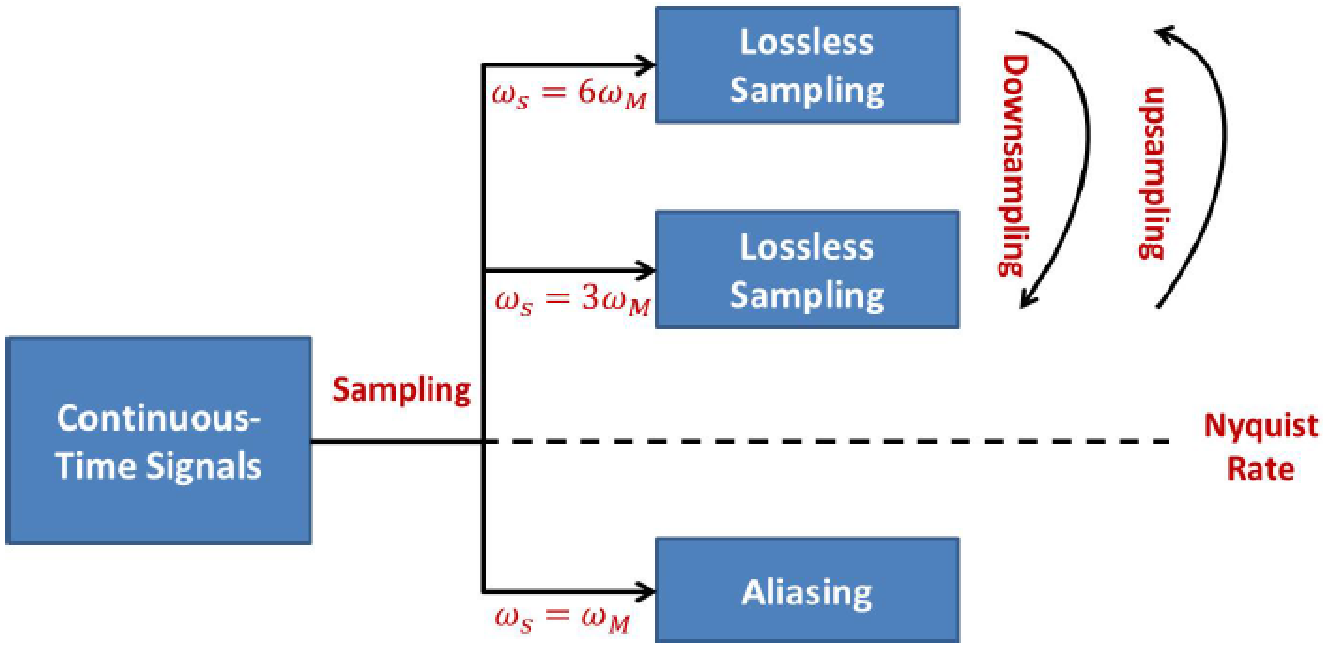
$$\boxed{\omega_s = \frac{2\pi}{N}}$$

Decimation:

$$\boxed{\omega_s = 2\pi}$$

Condition for Perfect Reconstruction: $N\omega_M < \pi$

Anything Else?

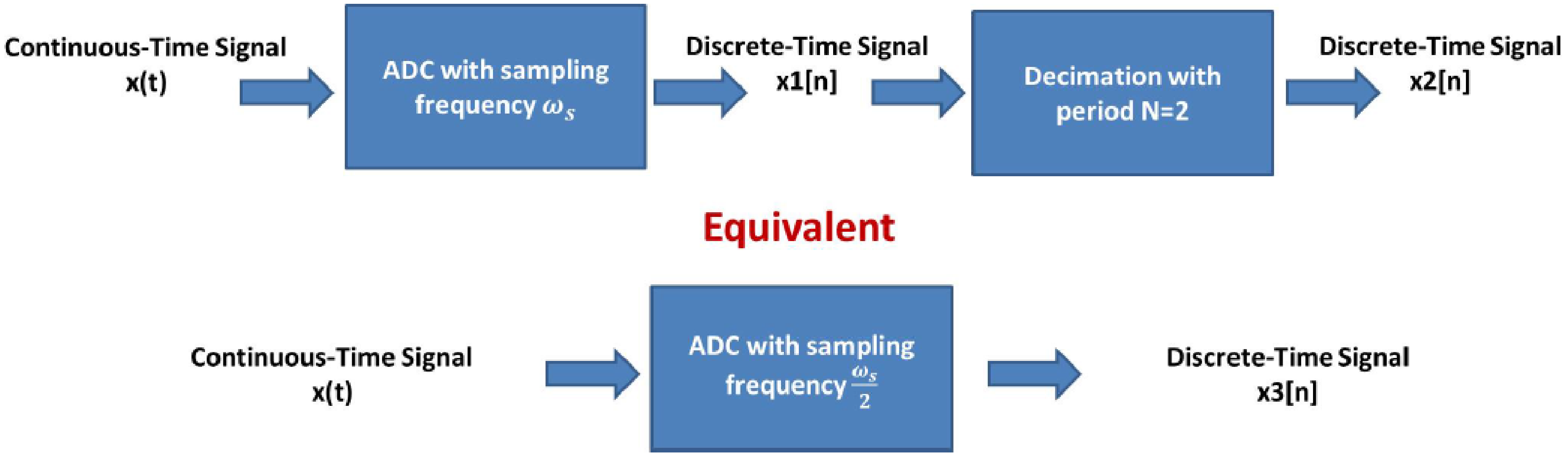


降采样率

- Downsampling: to reduce the sampling frequency (decimation)
- Upsampling: to generate a DT signal with higher sampling frequency
- As long as Nyquist rate is satisfied, the transform between low-sampling-frequency version and high-sampling-frequency versions is lossless.

Downsampling

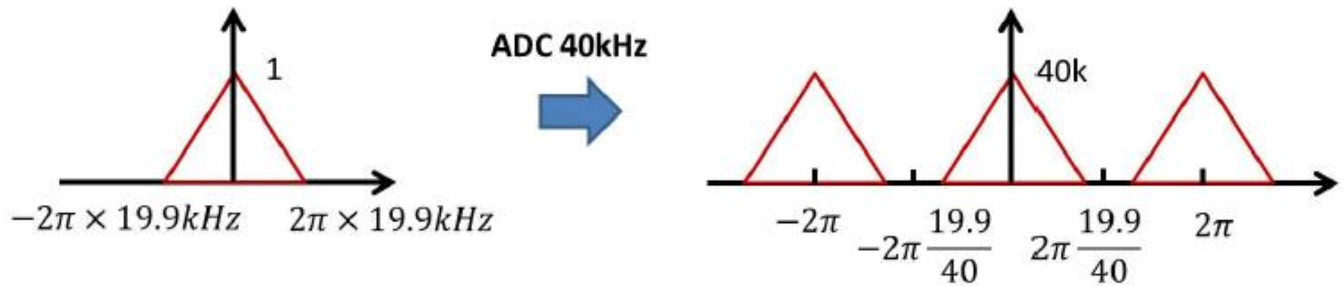
- Downsampling: a general procedure to reduce the sampling frequency



When do we need downsampling?

Downsampling Example (1/2)

- Suppose we have a clip of voice, $x(t)$, with bandwidth $=$19.9kHz
- It can be converted to DT signal with sampling frequency 40kHz, denoted as $x_1[n]$

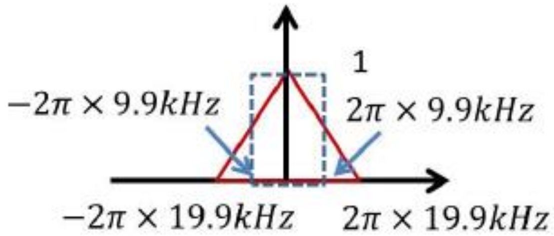


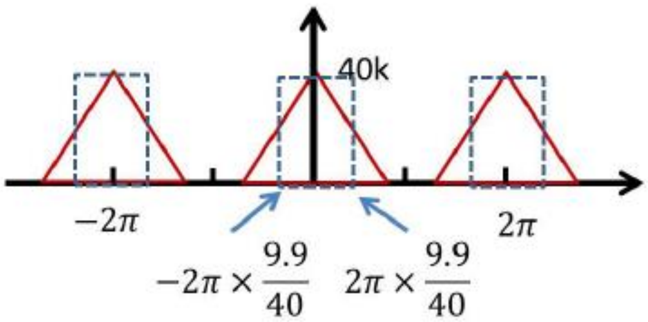
- Based on $x_1[n]$, if we want to save the voice information within 9.9kHz into another DT signal, what can we do?

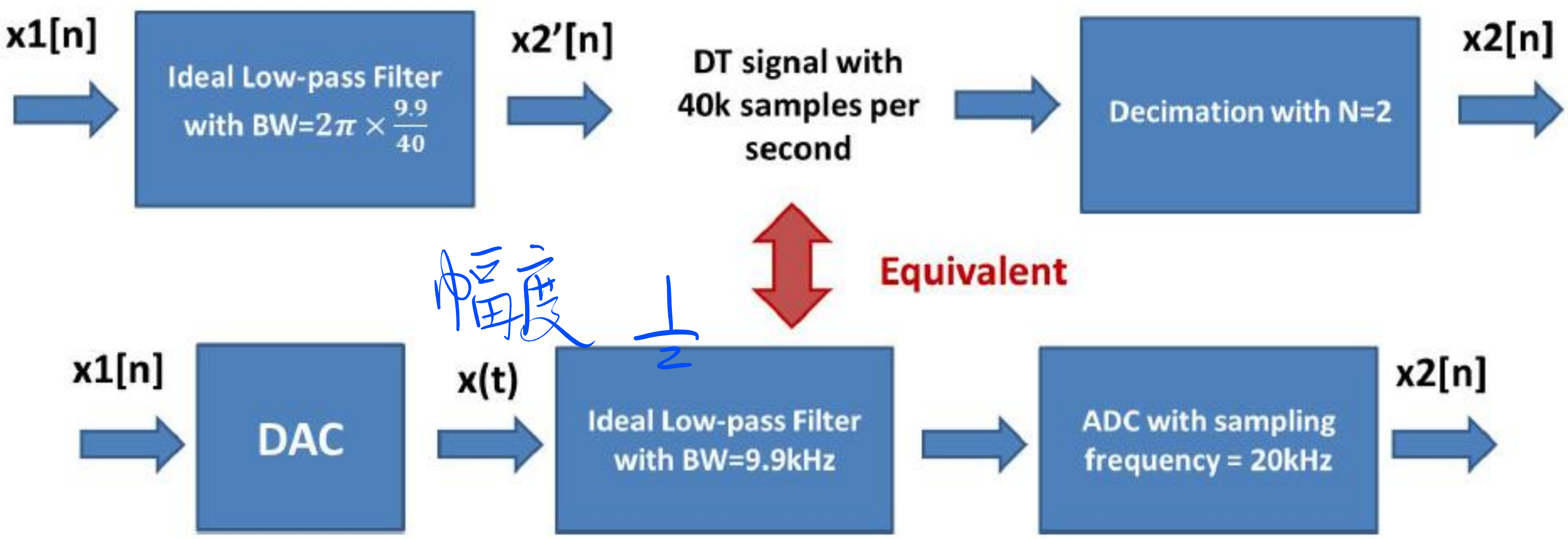
One Choice:

How can we generate x2[n] in discrete-time domain?

Downsampling Example (2/2)

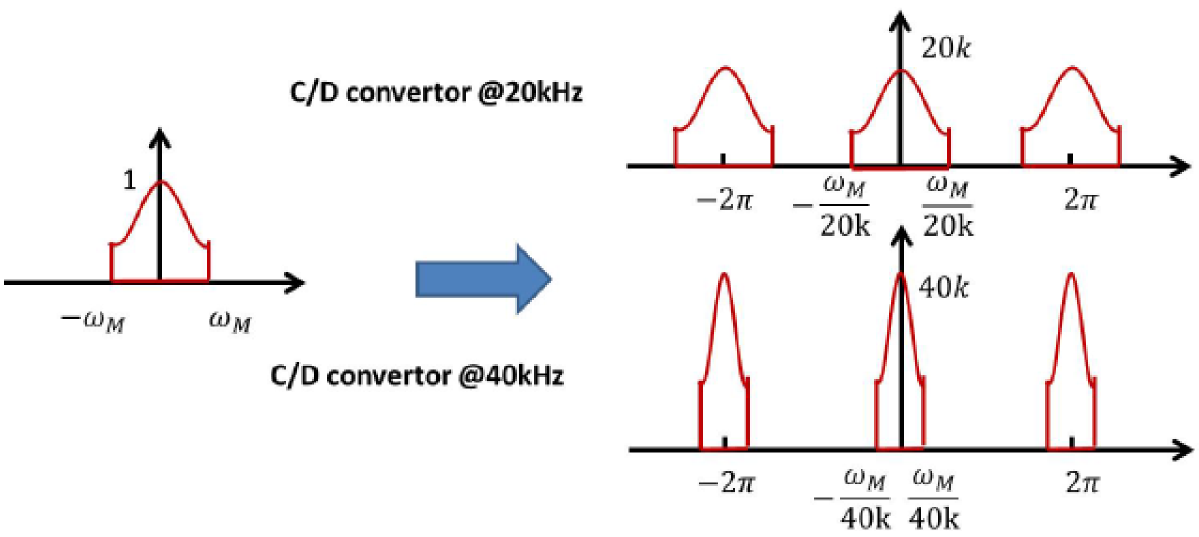






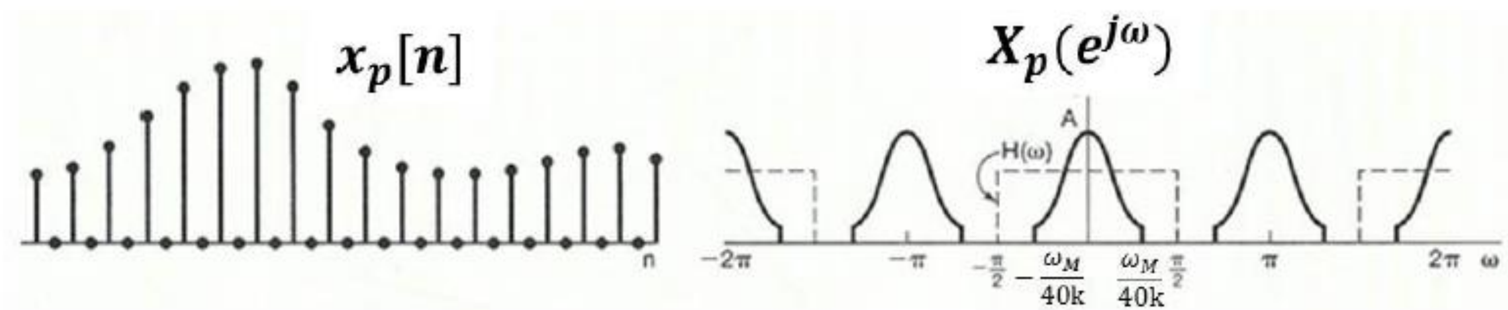
Upsampling

- Upsampling: a procedure to generate a sequence with higher sampling frequency
- Superpose the following two digital sound clips
  - Audio clip 1: Bandwidth= 19.9kHz, sampled at 40kHz
  - Audio clip 2: Bandwidth= 9.9kHz, sampled at 20kHz
- Double the sampling frequency of audio clip 2 (40kHz)
- How to do upsampling in discrete-time domain?



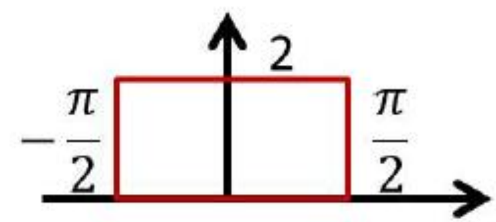
- Time expansion (Insert zeros):

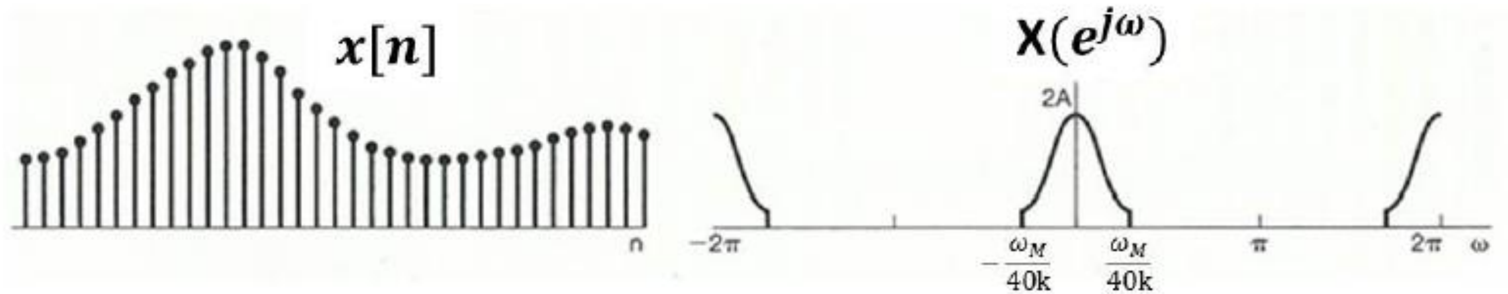
$$x_p[n] = x_{b(k)}[n] \longleftrightarrow X_p(e^{j\omega}) = X_b(e^{jk\omega})$$



- Low-pass filtering:

Ideal Low-Pass Filter:





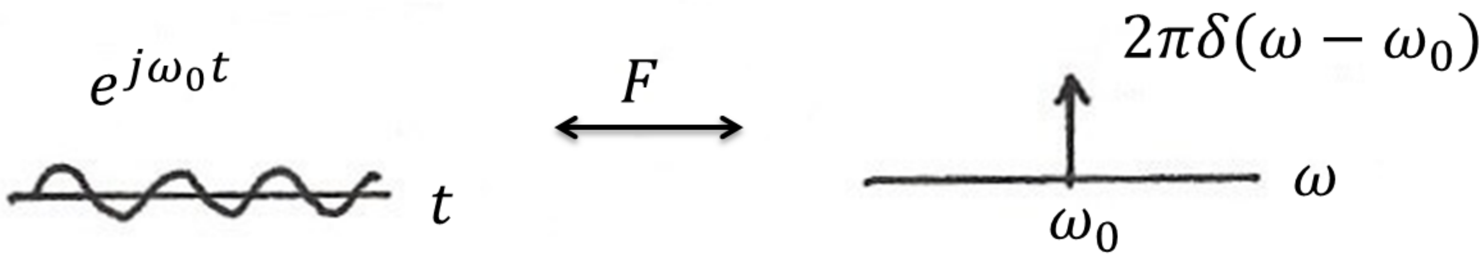
Chapter 8
Communication Systems

AM FM Angle

phase

freq

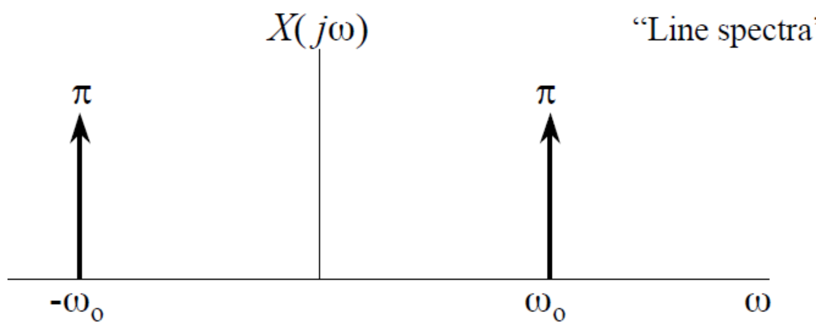
Review



$$x(t) = \cos\omega_0 t = \frac{1}{2}e^{j\omega_0 t} + \frac{1}{2}e^{-j\omega_0 t}$$

$$\updownarrow$$

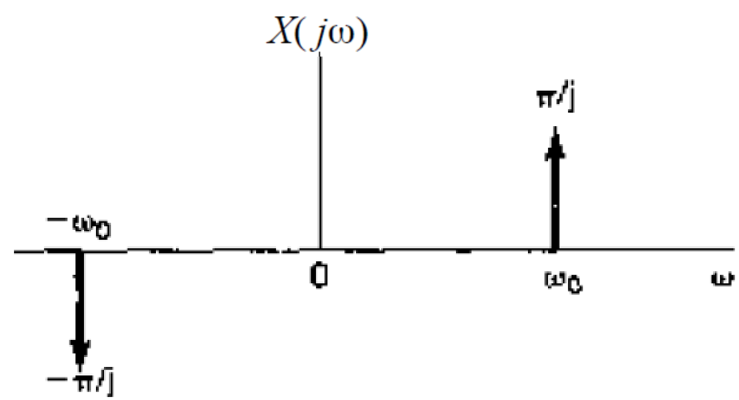
$$X(j\omega) = \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$



$$x(t) = \sin(\omega_0 t) = \frac{1}{2j}e^{j\omega_0 t} - \frac{1}{2j}e^{-j\omega_0 t}$$

$$\updownarrow$$

$$X(j\omega) = \frac{\pi}{j}\delta(\omega - \omega_0) - \frac{\pi}{j}\delta(\omega + \omega_0)$$



Communication Systems

s(t) → **Transmitter** → y(t) → **Channel** → **Receiver** → w(t)

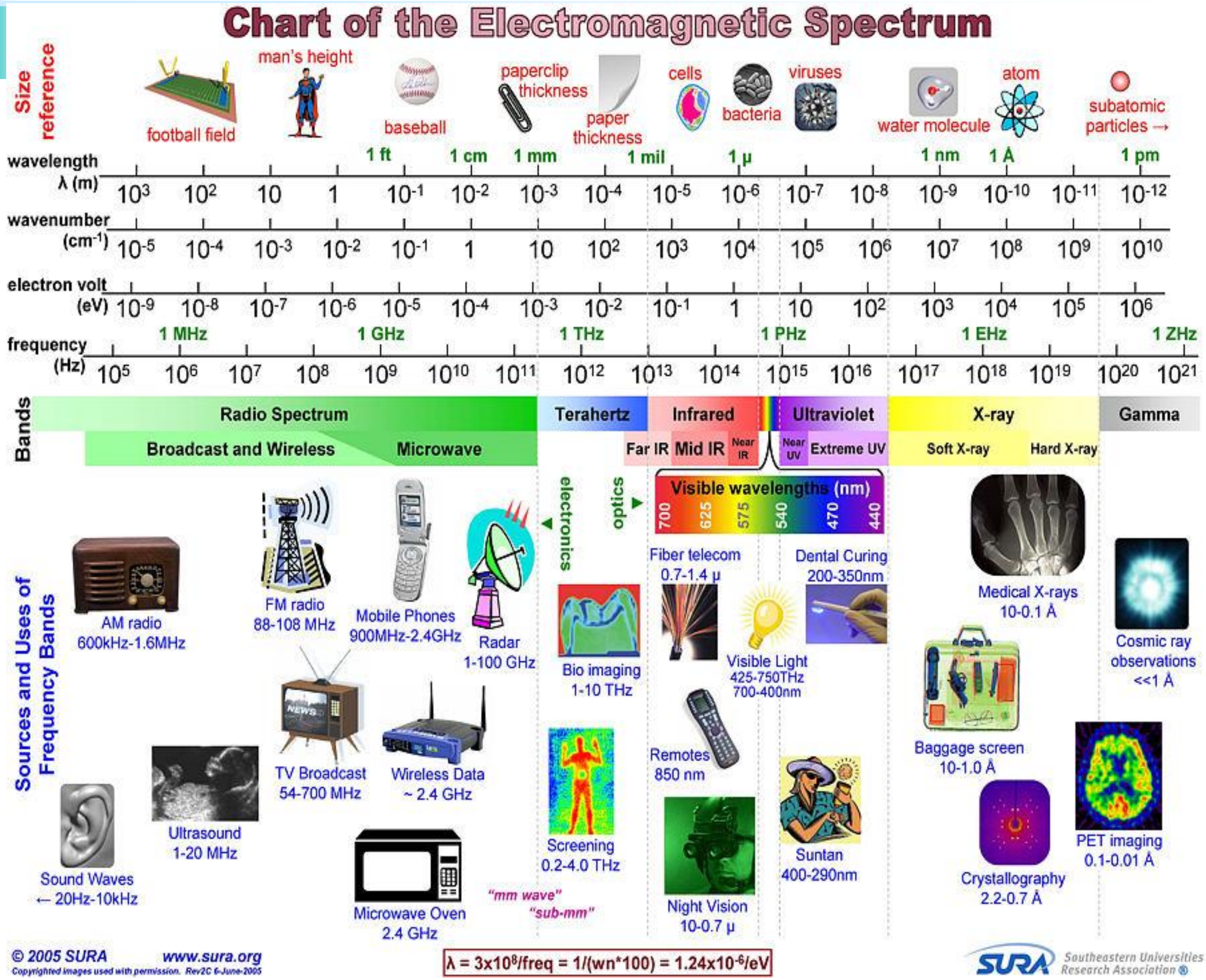
Message signal (speech, text, image, video)

Transmitted signal

Received signal

Recovery of message signal

Channel (media) Example

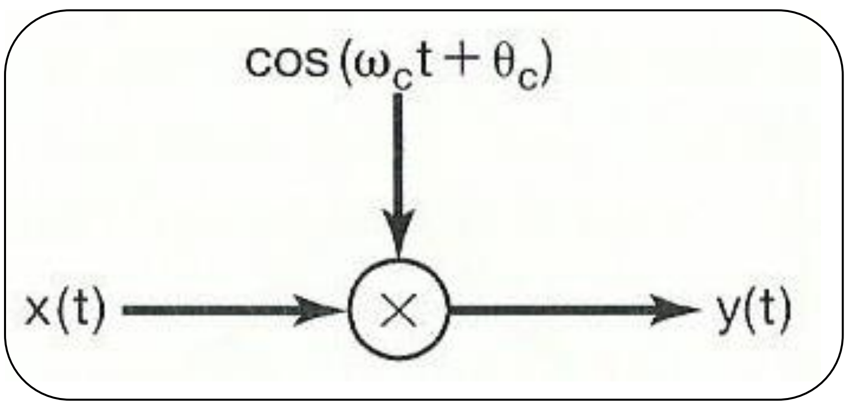


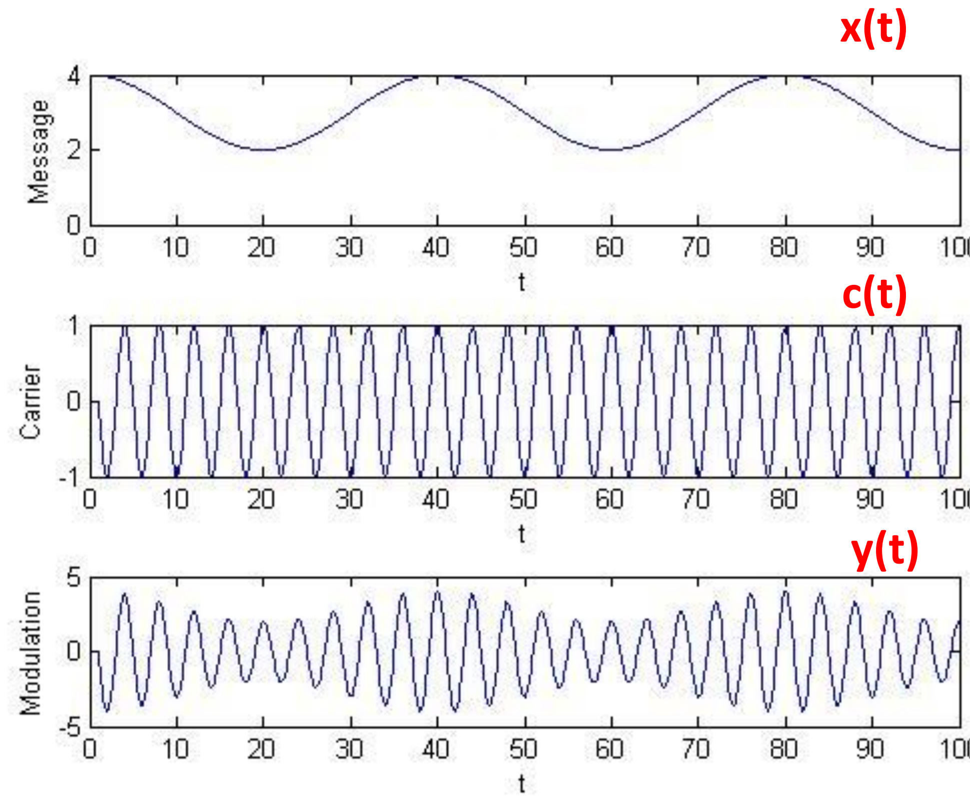
Amplitude modulation

- Modulation: the general process to embed an information-bearing signal into a second signal.

$$y(t) = x(t)c(t)$$

x(t): **m**odulating (or message) signal
c(t): **c**arrier signal
y(t): modulated signal





Amplitude modulation with a complex exponential carrier

ω_M : the highest frequency in x(t)

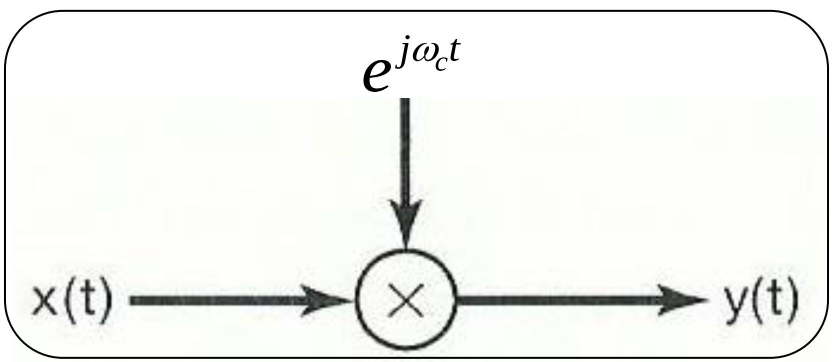
$$c(t) = e^{j(\omega_c t + \theta_c)}$$

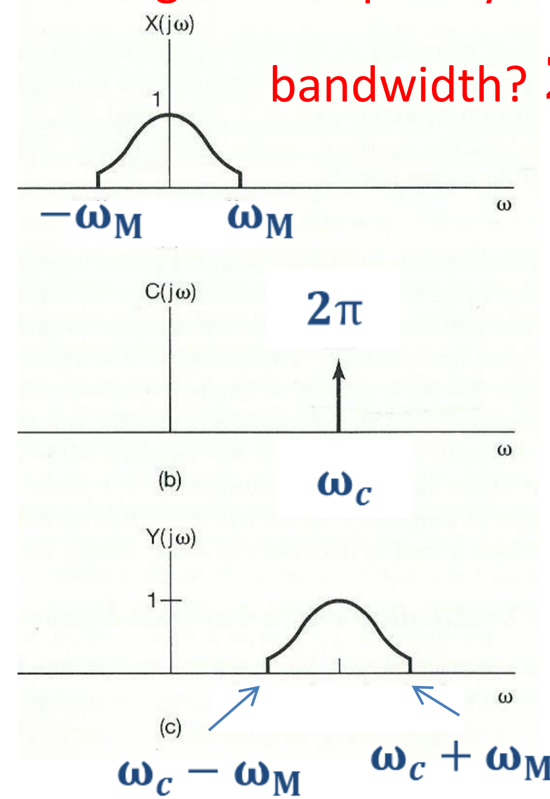
ω_c : carrier frequency

bandwidth? $2\omega_M$

First, we suppose $\theta_c = 0$

$$y(t) = x(t)c(t) = x(t)e^{j\omega_c t}$$

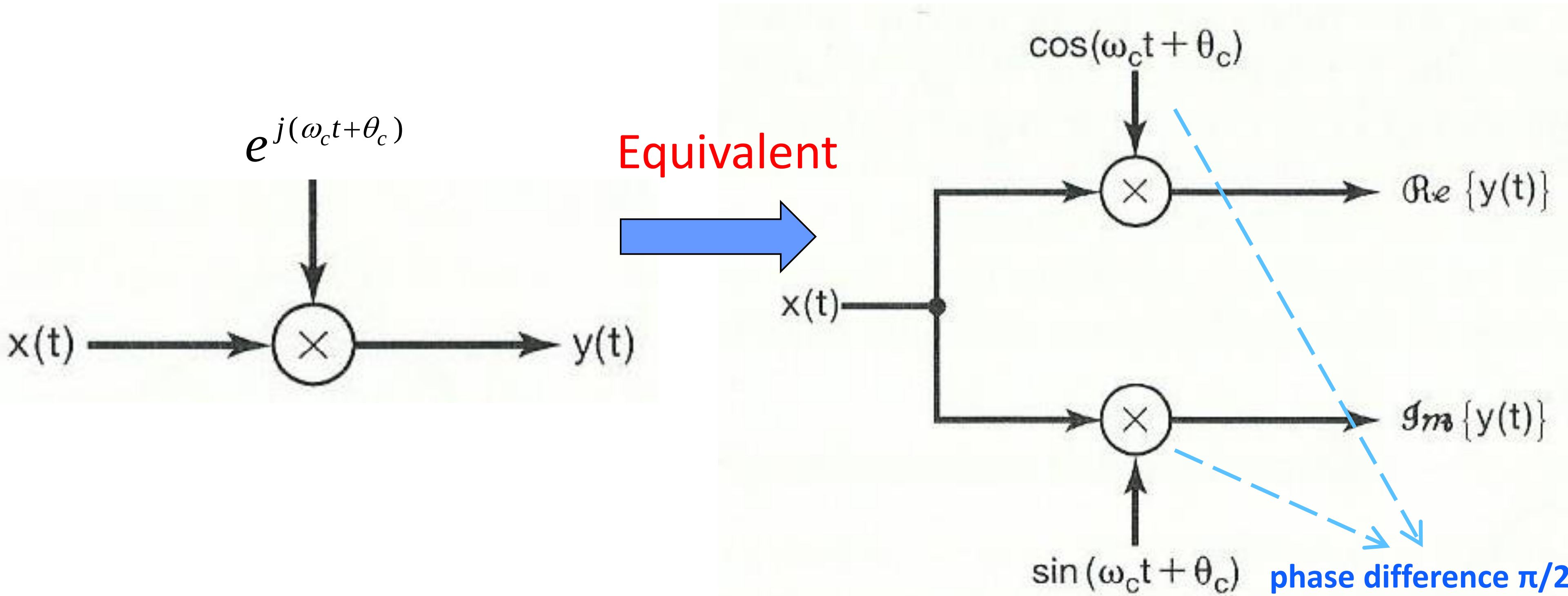




- The spectrum of the modulated output y(t) is that of modulating input x(t), shifted in frequency by amount of carrier frequency ω_c.
- How to recover the modulating input x(t) from modulated signal y(t)?

Amplitude modulation with a complex exponential carrier (cont.)

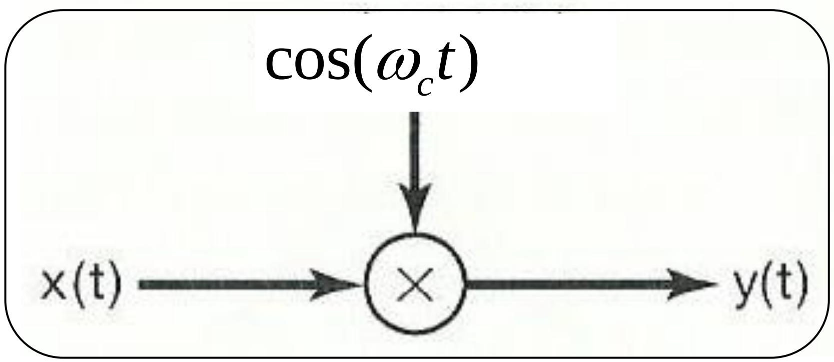
$$y(t) = x(t)e^{j\omega_c t} = x(t)\cos(\omega_c t) + jx(t)\sin(\omega_c t)$$

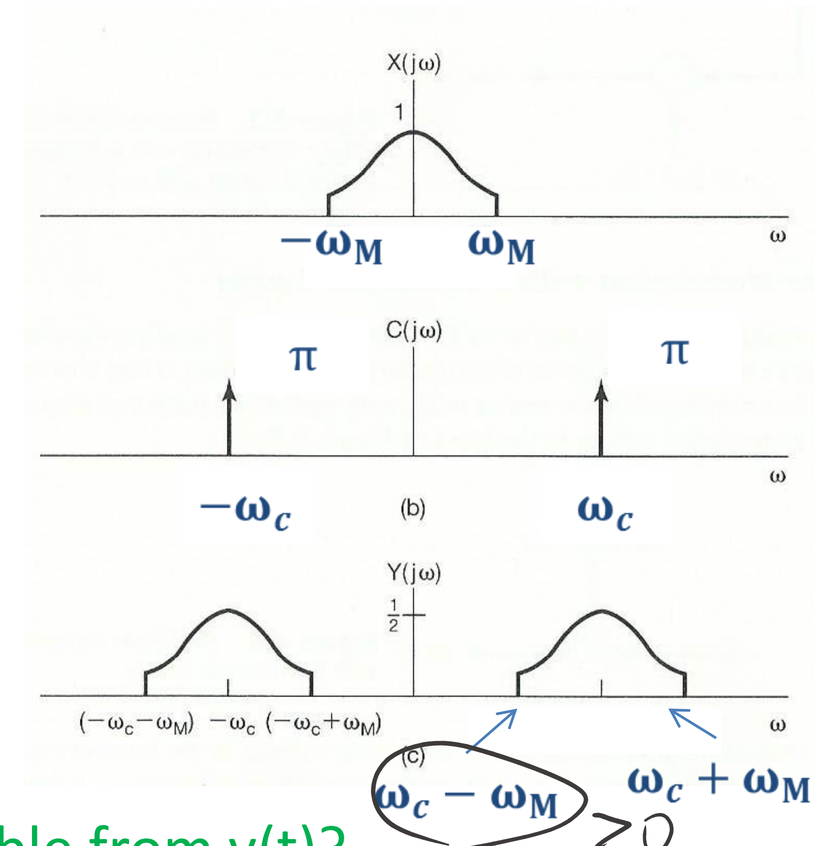


Amplitude modulation with a sinusoidal carrier

First, we suppose $\theta_c = 0$

$$y(t) = x(t)c(t) = x(t)\cos(\omega_c t)$$





What is required if x(t) is recoverable from y(t)?

Condition: $\omega_c > \omega_M$

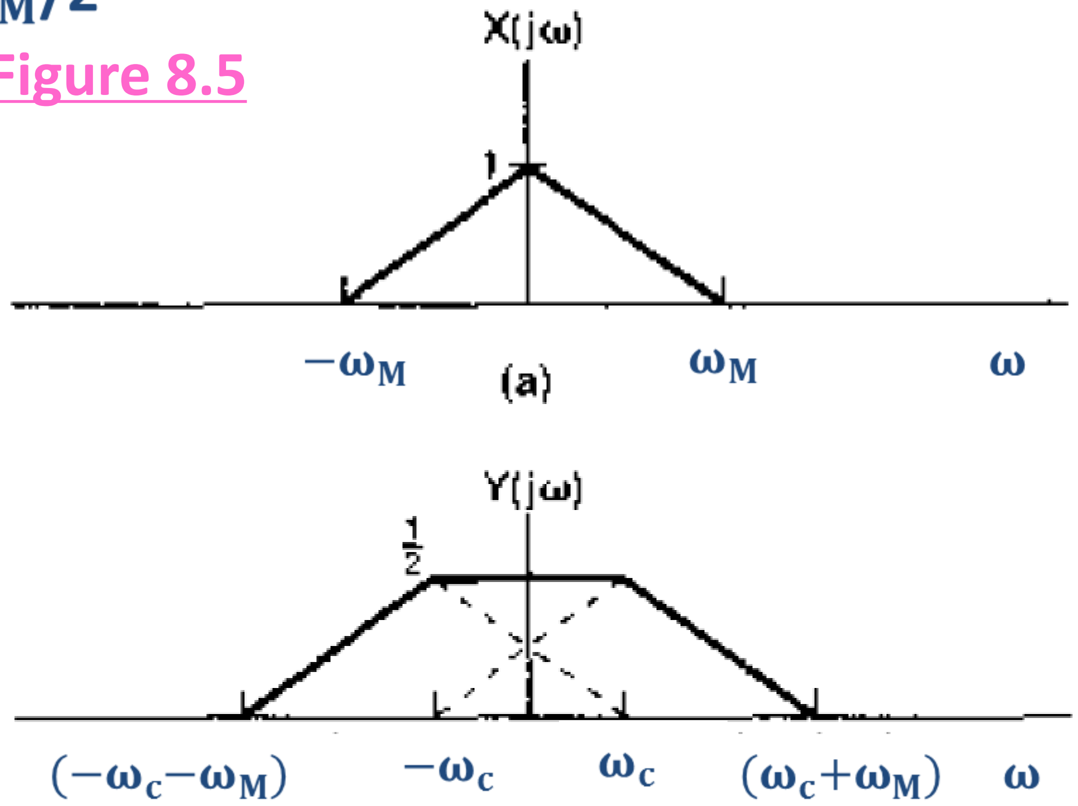
Amplitude modulation with a sinusoidal carrier (cont.)

What happens if $\omega_c < \omega_M$?

Example of an overlapping between the two replications of X($j\omega$)

$\omega_c = \omega_M/2$

Figure 8.5



Demodulation for sinusoidal AM

- Purpose: to recover the information-bearing signal x(t) from the modulated signal y(t).
- Two types of AM demodulation:
  - 同步解调
  - Synchronous demodulation, in which transmitter and receiver are synchronized in phase and frequency
  - Asynchronous demodulation
  - 异步解调

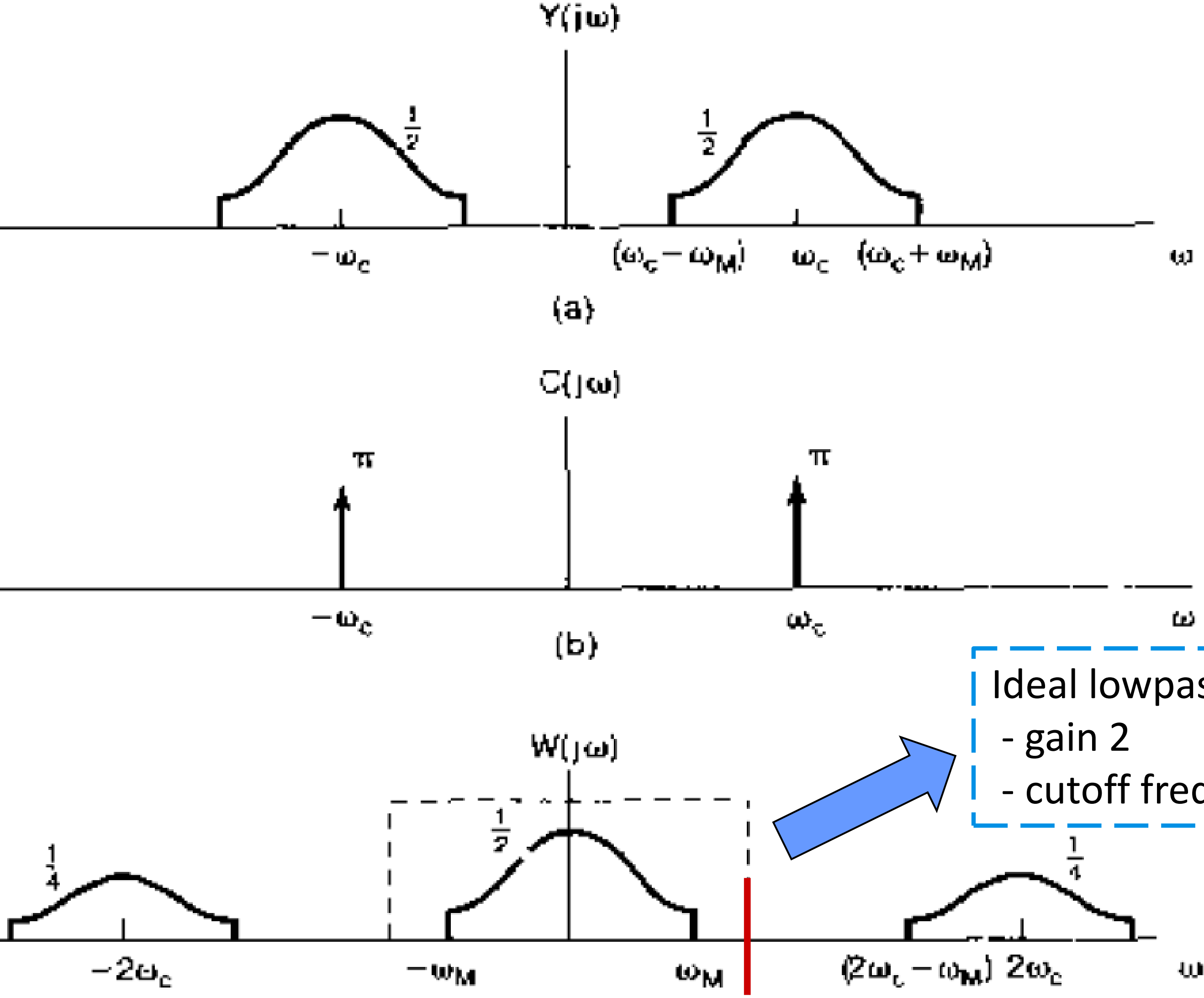
Synchronous demodulation

With sinusoidal carrier

$$y(t) = x(t)\cos(\omega_c t)$$

$$w(t) = y(t)\cos(\omega_c t)$$

$w(t)$: demodulated signal



Ideal lowpass filtering
- gain 2
- cutoff frequency > ω_M and < $2\omega_c - \omega_M$

Synchronous demodulation (cont.)

- Mathematically,

AM modulation → $$y(t) = x(t)\cos(\omega_c t)$$

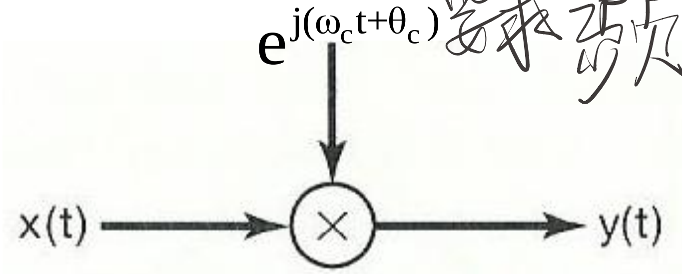
AM demodulation → $$\begin{aligned} w(t) &= y(t)\cos(\omega_c t) = x(t)\cos^2(\omega_c t) \\ &= x(t)[\frac{1}{2} + \frac{1}{2}\cos(2\omega_c t)] \\ &\boxed{= \frac{1}{2}x(t)} + \frac{1}{2}x(t)\cos(2\omega_c t) \end{aligned}$$

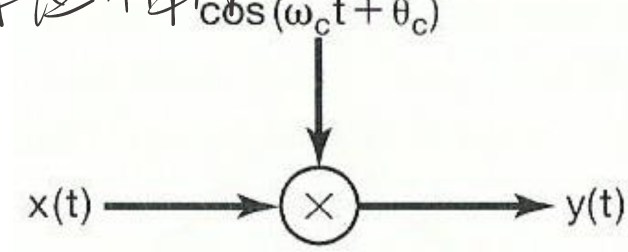
Synchronous demodulation (cont.)

Complex exponential carrier: **Sinusoidal carrier:**

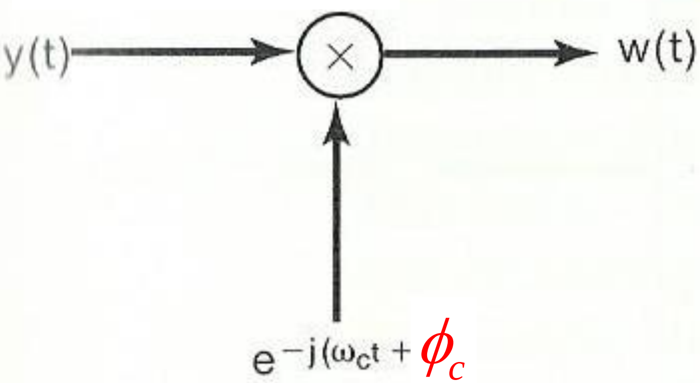
AM modulation

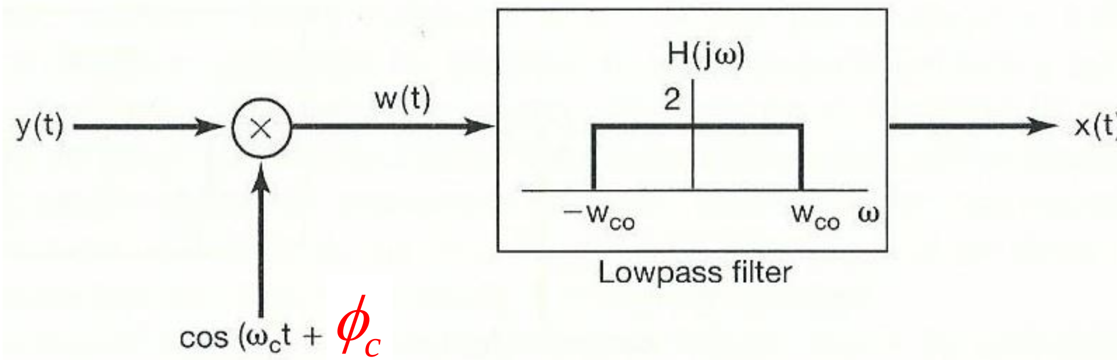
要求频率和相位相同





De-modulation





$$y(t)=e^{j(\omega_c t+\theta_c)}x(t)$$

$$w(t)=e^{-j(\omega_c t+\phi_c)}y(t)$$

$$=e^{j(\theta_c-\phi_c)}x(t)$$

$$w(t)=x(t)\cos(\omega_c t+\theta_c)\cos(\omega_c t+\phi_c)$$

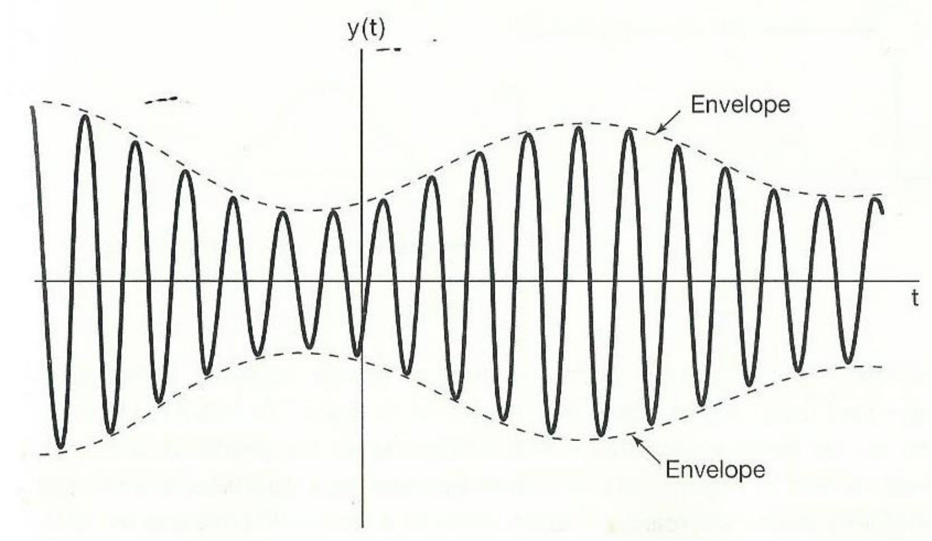
$$\boxed{=\frac{1}{2}\cos(\theta_c-\phi_c)x(t)}+\frac{1}{2}x(t)\cos(2\omega_c t+\theta_c+\phi_c)$$

What is the cost ?

Asynchronous demodulation

- Avoid the need for synchronization between modulator and demodulator.

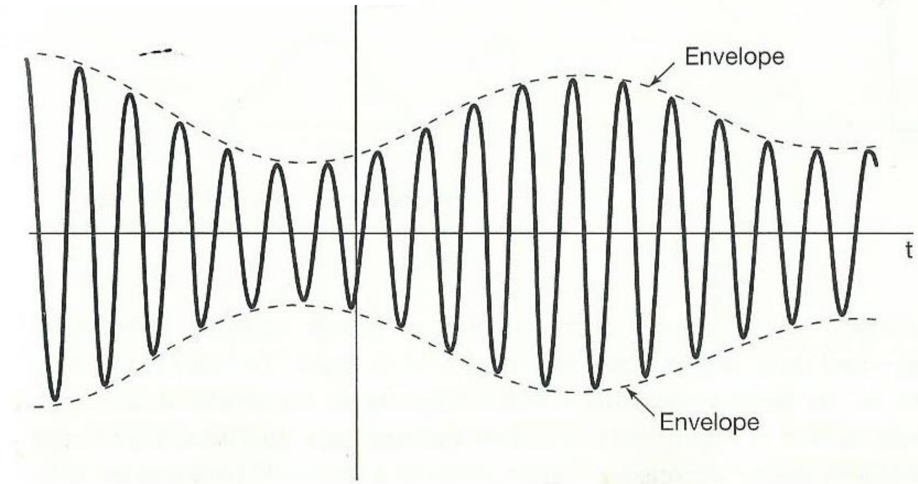
An example of modulated signal y(t), when x(t) is positive, and $\omega_c > \omega_M$.



Envelope signal, a smooth curve connecting the peaks, approaches to the modulating signal x(t).

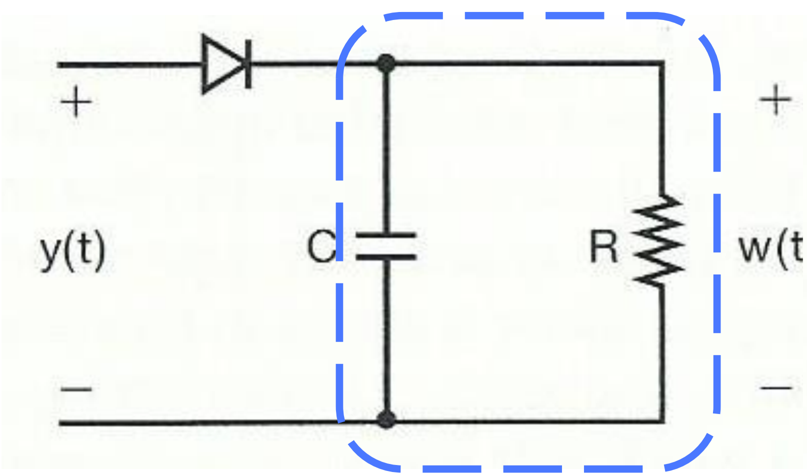
Envelope detector

Modulated signal

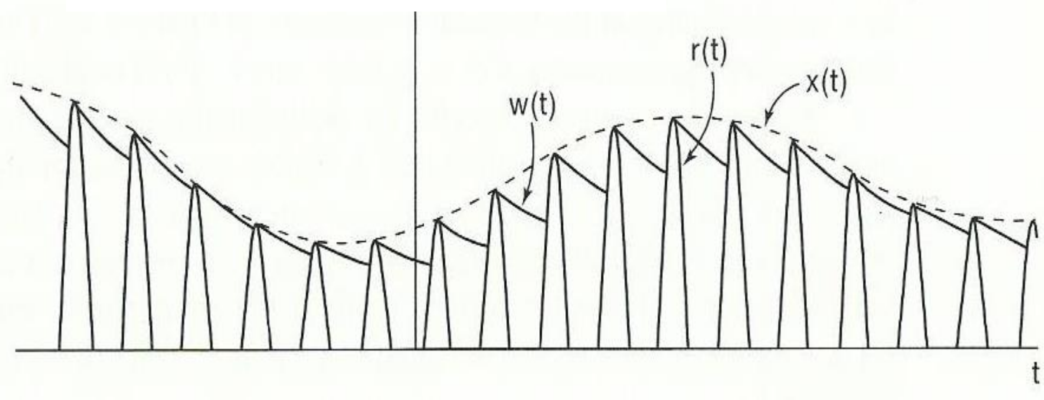


Full wave or half-wave envelope detector?

Lowpass filtering

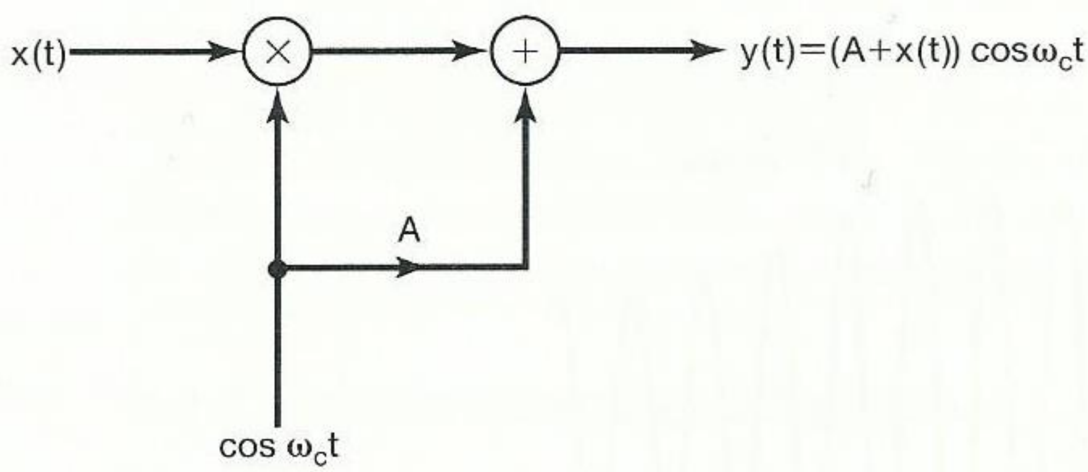


Envelope detector output



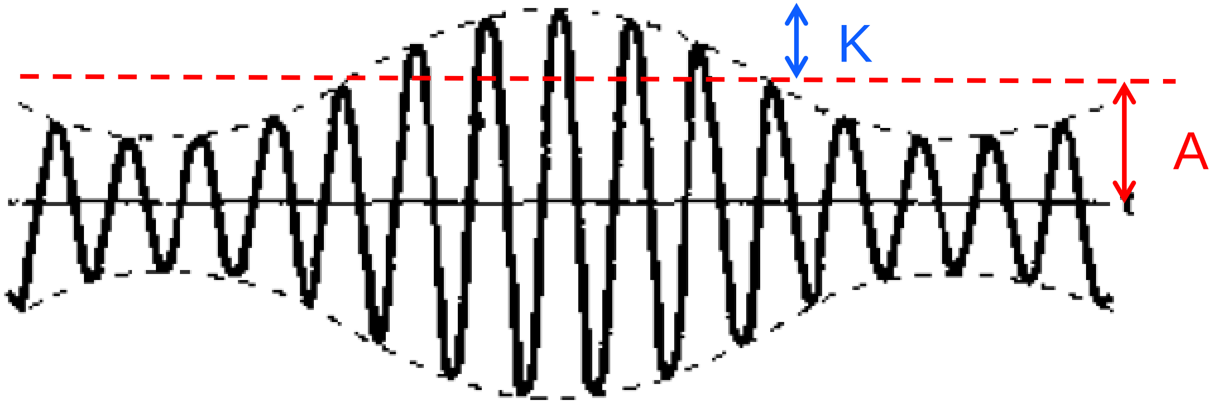
AM in an asynchronous system

- Two important assumptions:
  - ω_c is higher than ω_M $\quad w_c >> \omega_m$
  - x(t) is always positive



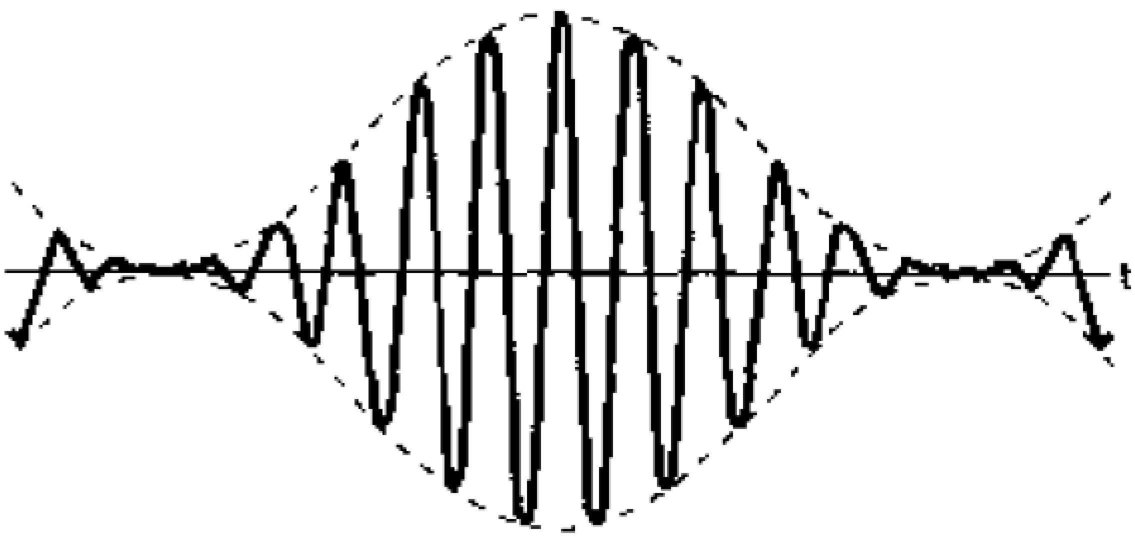
1) Add a constant A to make sure [A+x(t)] is positive.
Let K be the maximum amplitude of x(t), or $|x(t)| \leq K$. We need to have A>K.

2) Define K/A as modulation index *m*, which is from 0 to 1.



(a)

m=0.5



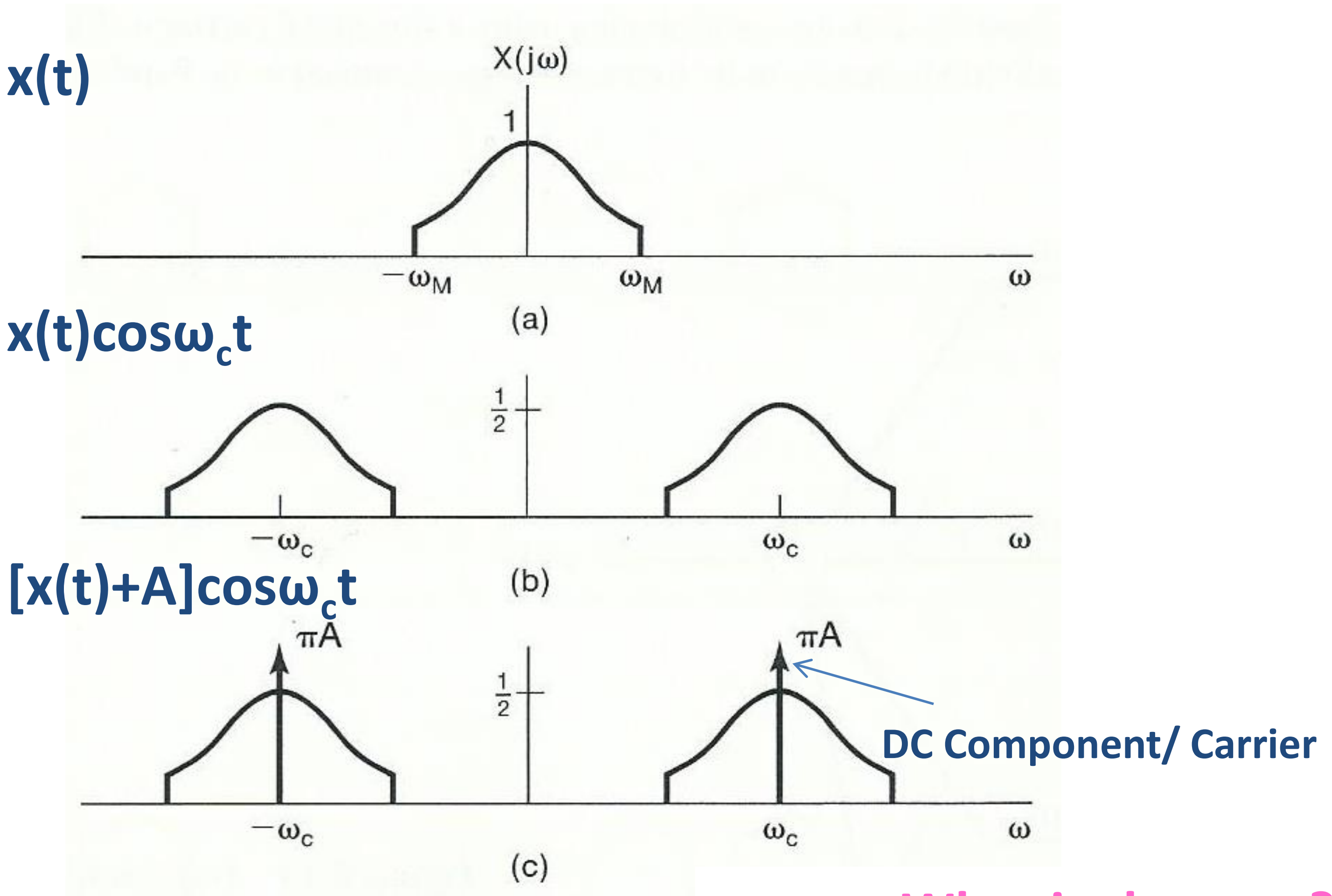
(b)

m=1

What about *m*=0?

AM in an asynchronous system (cont.)

Spectrum of modulated signal in an asynchronous system

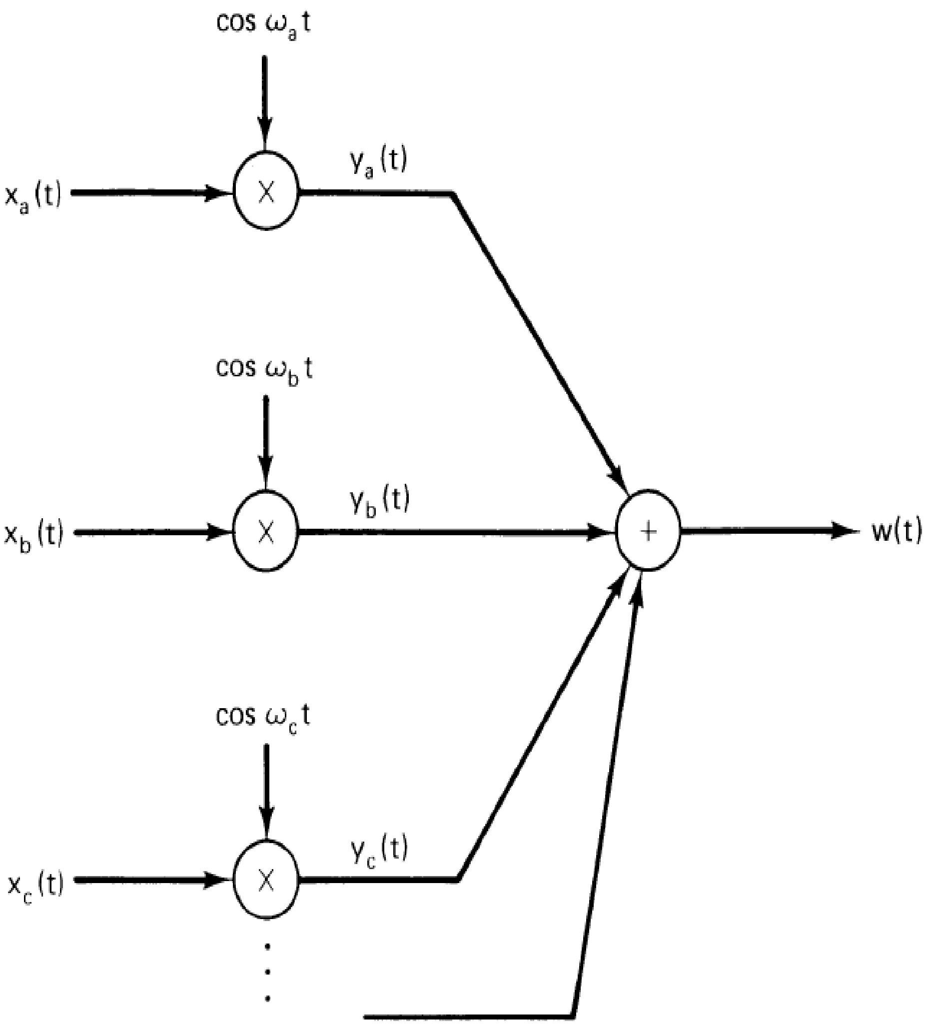


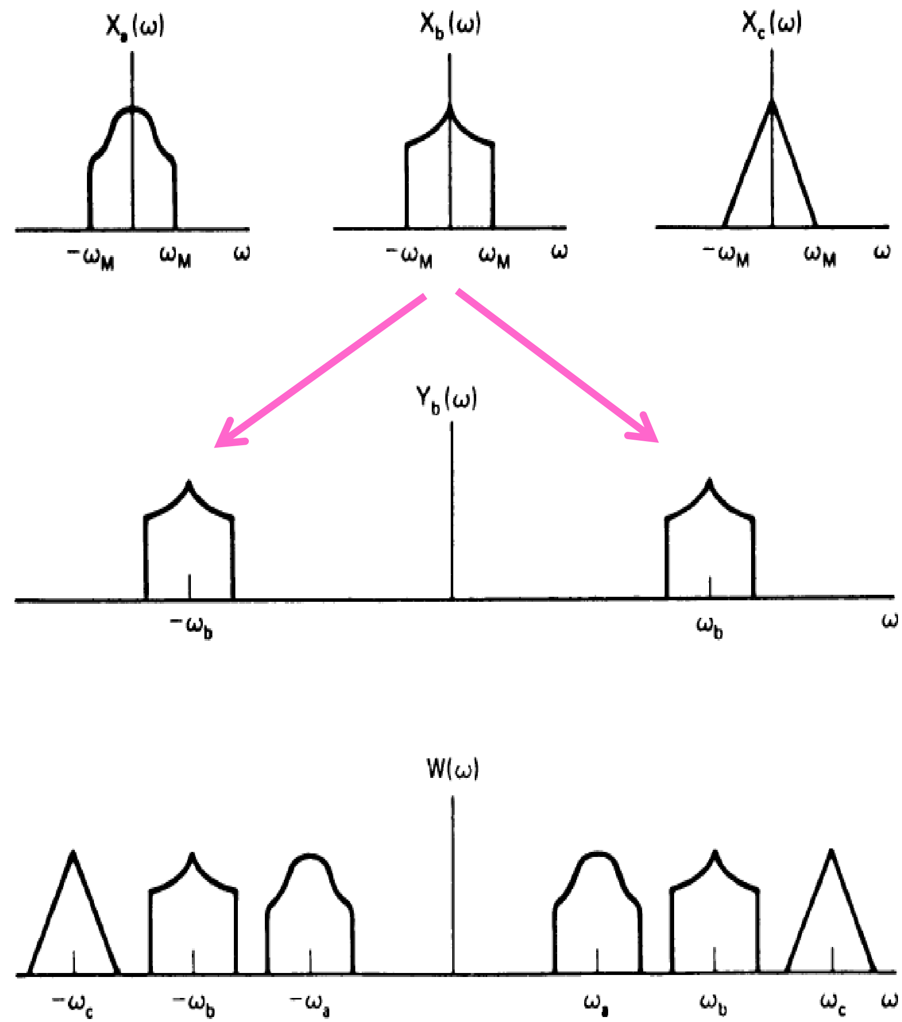
What is the cost ?

Frequency-division multiplexing (FDM)

- Systems for transmitting signals provide more bandwidth than is required for one signal.
  - E.g., speech signal → 20 ~ 20 kHz
    microwave channel → 300 MHz ~ 300 GHz
    satellite link → a few hundred MHz ~ 40 GHz
    (more in Fig. 8.18)
- Different modulating signals (e.g., speech), which are overlapping in frequency, can have their spectra shifted (e.g., by sinusoidal AM) without overlapping, so they can be transmitted simultaneously over a single wideband channel.

FDM (cont.)

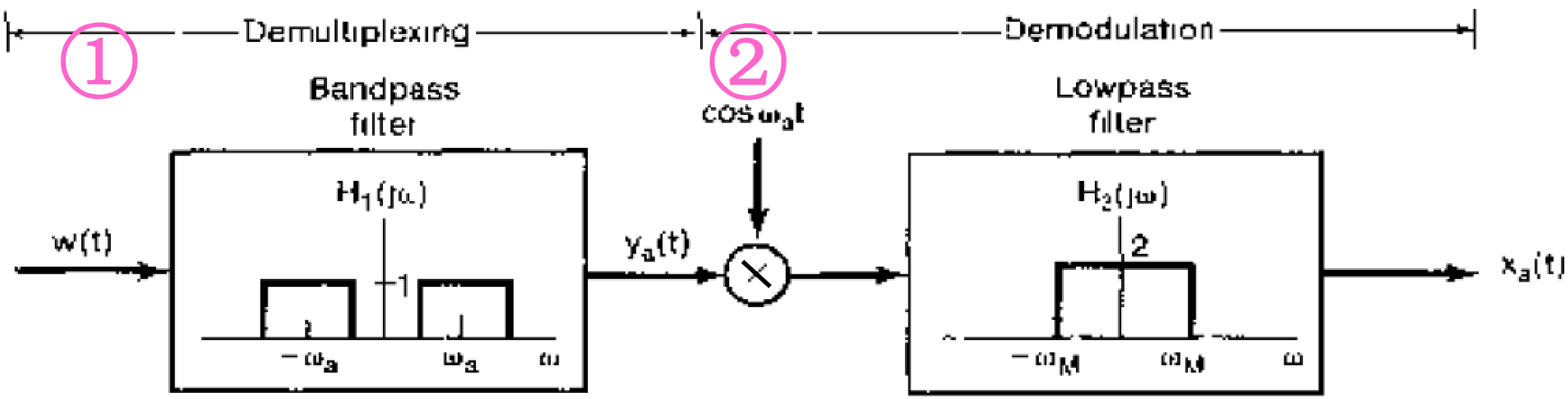

cos ω_a t
x_a(t)
y_a(t)
cos ω_b t
x_b(t)
y_b(t)
w(t)
cos ω_c t
x_c(t)
y_c(t)

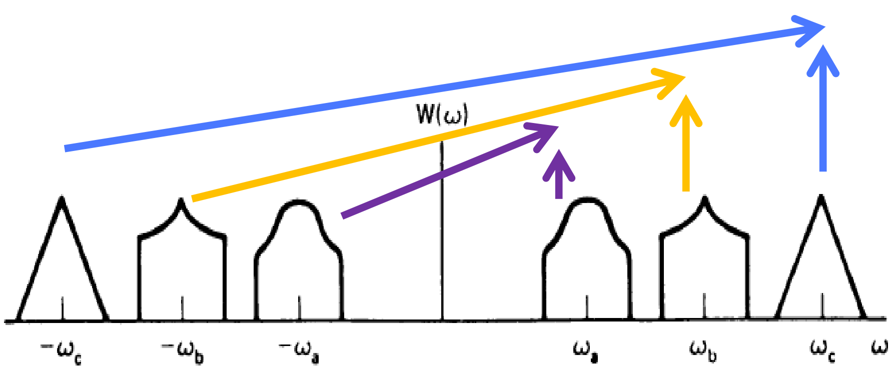

$X_a(\omega)$
$X_b(\omega)$
$X_c(\omega)$
$-\omega_M$
ω_M
ω
$Y_b(\omega)$
$-\omega_b$
ω_b
$W(\omega)$
$-\omega_c$
$-\omega_a$
ω_a
ω_c

FDM demultiplexing and demodulation

① bandpass filtering to have the modulated signal from one channel

② demodulation to recover the modulating signal



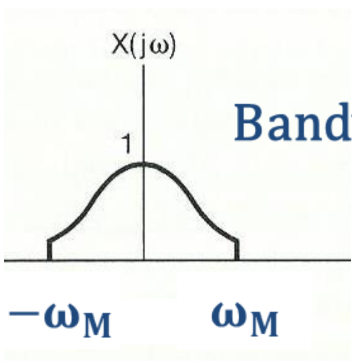


Occupy twice the original bandwidth

→ insufficient use of bandwidth

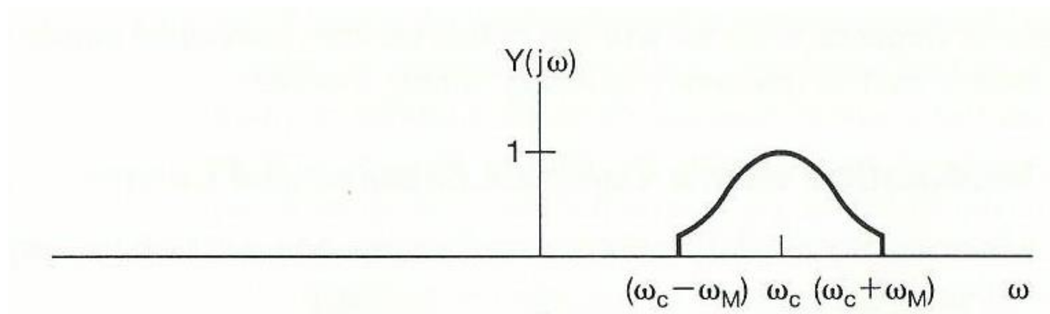
Single-sideband (SSB) sinusoidal AM

Occupied bandwidth:

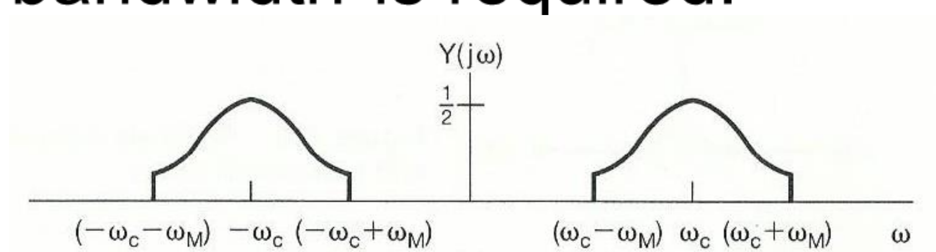


Bandwidth: $2\ \omega_M$

- With exponential carrier, the bandwidth is still $2\ \omega_M$

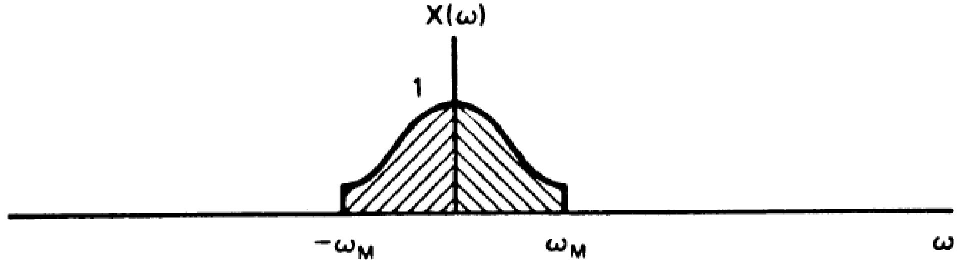


- With sinusoidal carrier, twice bandwidth is required.



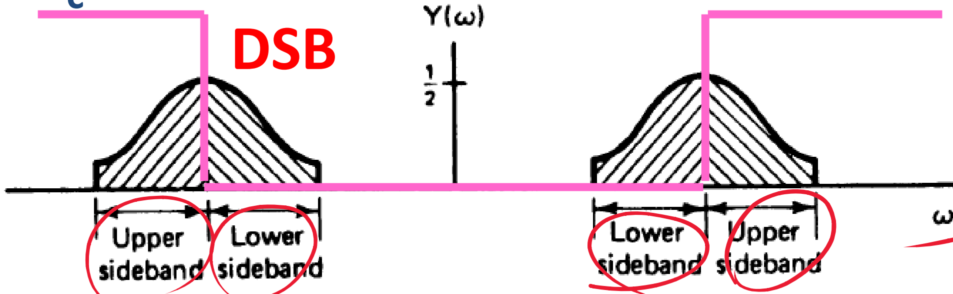
SSB sinusoidal AM (cont.)

x(t)



x(t)cosω$_c$t **Double-sideband** Ideal Highpass filter

DSB

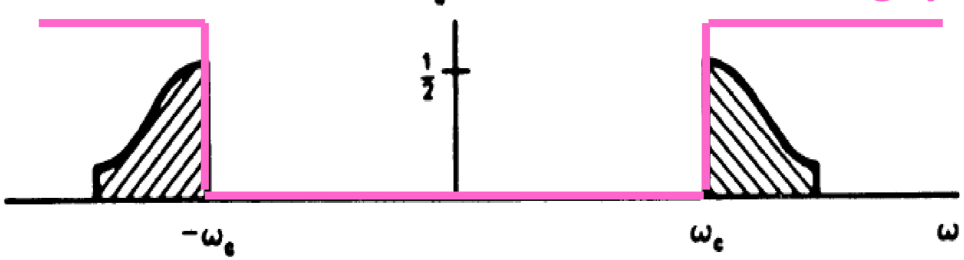


卷积使之对称

<u>Observation:</u> x(t) can be recovered if two upper (or lower) sidebands are retained.

SSB

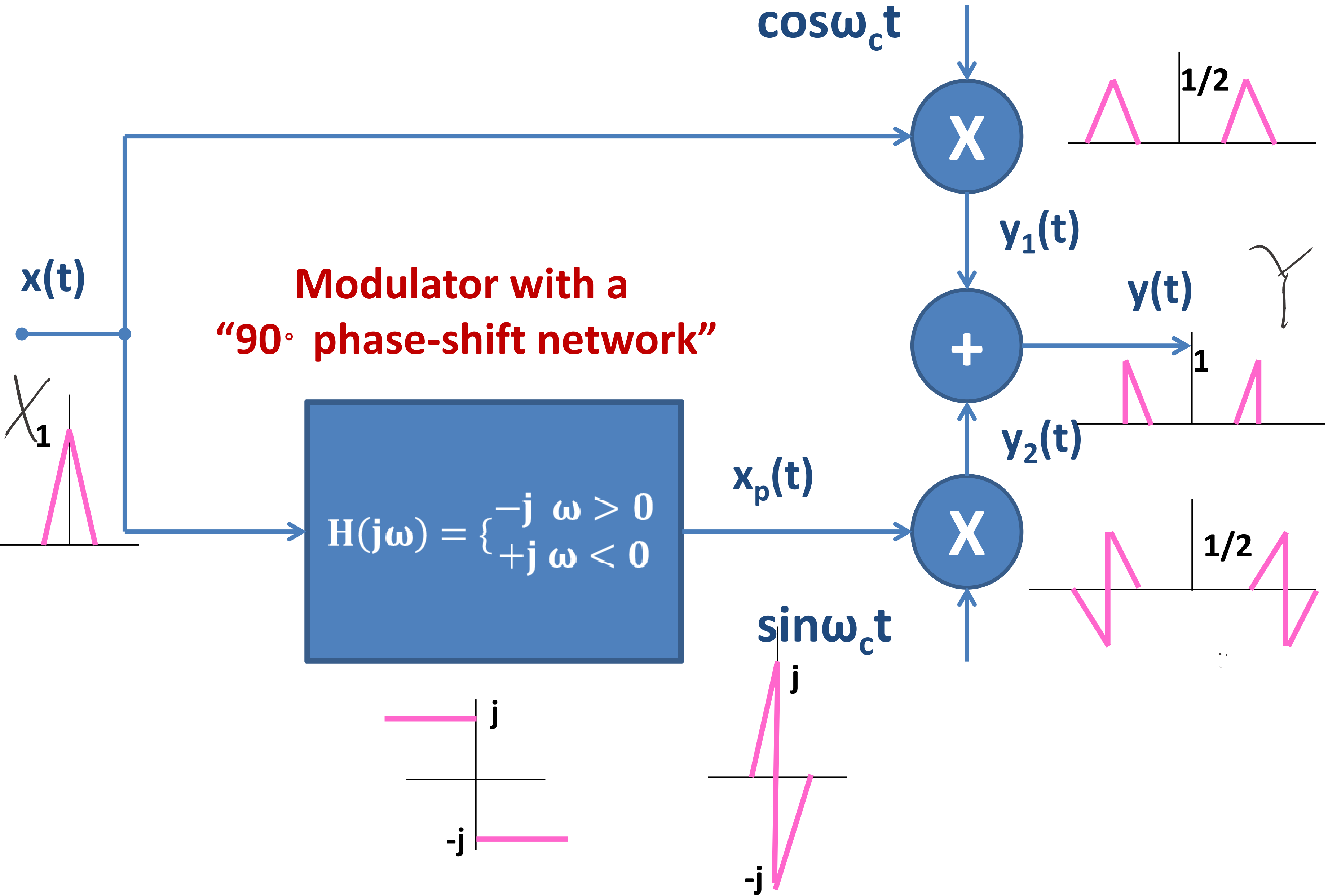
Ideal Highpass filter



Higher power and spectrum efficiency

What is the cost ?

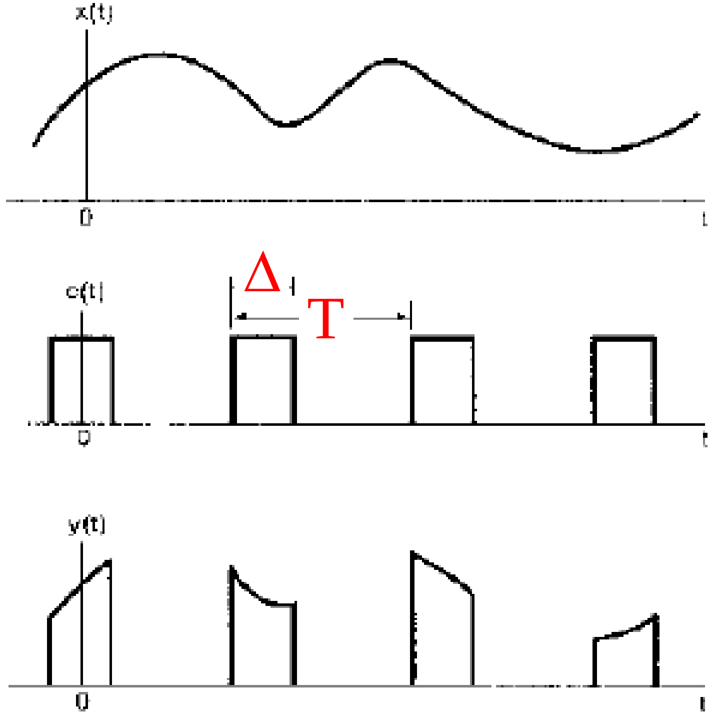
SSB sinusoidal AM (cont.)



Amplitude modulation with a pulse train

- Carrier signal could be a sinusoidal signal, or a pulse train.

Figure 8.23



$$X(j\omega)$$

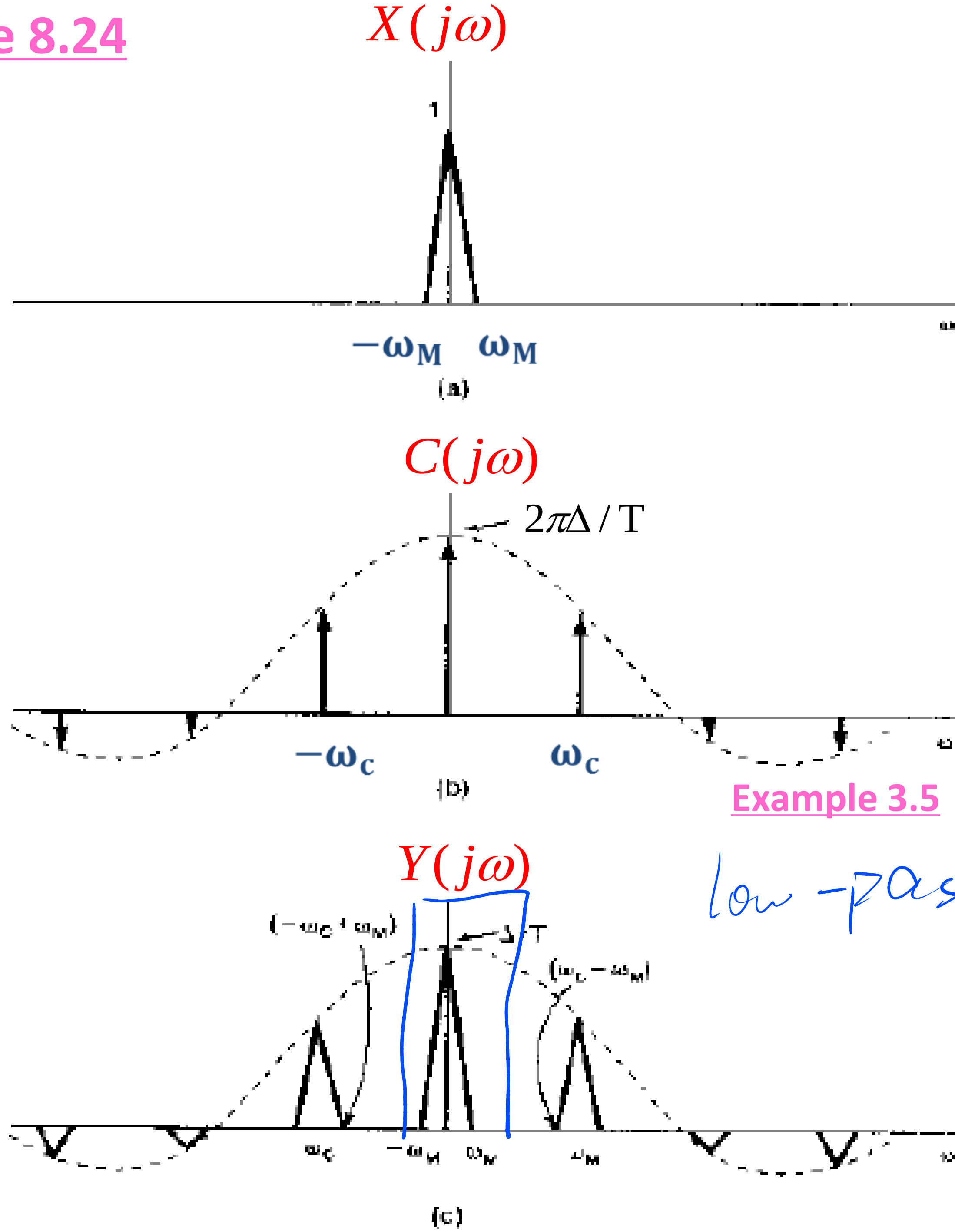
$$C(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_c)$$

$$a_k = \frac{\sin(k\omega_c \Delta / 2)}{\pi k}$$

Read Example 3.5

$$Y(j\omega) = \sum_{k=-\infty}^{\infty} a_k X(j(\omega - k\omega_c))$$

Figure 8.24



Example 3.5

When $\omega_c > 2\omega_M$, $X(j\,\omega)$ can be recovered by lowpass filtering.

Note: similar to the condition in sampling.

low-pass $\Rightarrow X(j\omega)$

Summary

- Meaning of amplitude modulation
  - with a complex exponential carrier
  - with a sinusoidal carrier
- Demodulation for sinusoidal AM
  - Synchronous demodulation
  - Asynchronous demodulation, and its two important assumptions
- Frequency-division multiplexing (FDM)